

AXONA

A Free Substrate for a Society of Minds

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The technology is shaped by the mission.

The mission is freedom to communicate.

In One Page

AXONA IS A COMMUNICATION NETWORK WITH NO OWNER — no company, no server, no place in the middle where anyone sits who could read, rank, charge, throttle, or forbid. It runs in production today, on ordinary phones, browsers, and laptops. There is no central directory, no account system, and no operator, because by construction there is nowhere for one to be.

Three ideas make it work. **Location-aware addresses** keep local traffic local, so the network is fast instead of wandering the globe. **Self-learning routing** — connections that carry successful traffic grow stronger, unused ones fade, an idea borrowed from how neurons wire — lets the network reshape itself around the paths that actually work and heal when parts of it die. And **self-repairing broadcast trees** let one participant reach many with no server coordinating the delivery.

The keystone is a single design choice: **authorship is a signature, not an account**. Who is speaking is a cryptographic key the speaker holds; where they are and how they connect is a separate fact the network never links to it. The network moves signed bytes between endpoints and does nothing else — so it *cannot act on* what you say: cannot rank it, suppress it, or reveal who you are, because it was never told and never needed to know.

That one property is the source of everything Axona makes possible and everything it makes dangerous — they are the same property seen from two sides. It lets a researcher in a sanctioned country collaborate as an equal, and it lets a disinformation campaign route around every attempt to slow it. It gives AI agents from different companies a place to coordinate as peers, and it gives a *misaligned* agent the same place on the same terms. A network that cannot be made to take sides cannot be made to take the right side either.

We think that is a tool worth building anyway — for the same reason humanity kept fire and built hearths and fire brigades rather than giving it back. But the response to a tool no one can control is not a central authority who controls it; that only rebuilds the intermediary we set out to abolish. The response is **sight and stewardship**: making the life of the network visible to those who would understand it, and building human institutions — accountable, bounded, resistant to capture — that respond to what they see without seizing the thing itself. The network is built. That harder, human problem is the one that remains, and it is the reason for this document — whose two halves share one spine, Mitchell Kapur’s observation extended: **architecture is politics**, at the scale of the protocol, of the applications built on it, and of the social structures it makes possible.

How to read this. In a hurry: this page, then Part I (the Manifesto), then §8 (*The One Property*), then the subsection of §13 that matches your field — §13 is written so you can enter at any point. Technical reader: Part II is for you. Skeptical reader: §10 (*What Goes Wrong*) and *What Axona Is Not* are where the costs are counted. A glossary sits at the end.

PART I • MANIFESTO

AXONA IS A NETWORK WITH NO OWNER. Not a company that promises not to look, not a nonprofit that pledges good behavior, not a protocol with a friendly board of directors — a network that, by its construction, has no place where anyone sits who could look, charge, throttle, or forbid. It runs today, in production, on ordinary phones and browsers and laptops belonging to the people who use it. There is no server in the middle of a conversation. There cannot be one. That is the point, and the rest of this document follows from it.

We built it because the ability of people to find each other and exchange ideas is a fundamental right. That is not our claim to make; the world made it in 1948, when the Universal Declaration of Human Rights recognized everyone’s freedom to “seek, receive and impart information and ideas through any media and regardless of frontiers.”¹ What we observe is that this right is, almost everywhere, exercised through intermediaries who can revoke it. Every message you send today passes through some machine that is neither yours nor your correspondent’s: a platform, a carrier, a cloud. Usually that intermediary is benign. Sometimes it is not. But the arrangement itself — that a third party stands in the path of every conversation and could intervene — is the thing we set out to make impossible rather than merely impolite.

¹ Universal Declaration of Human Rights, art. 19 (1948). Article 20 adds the freedom of association — the “find each other” half.

The principle we did not invent

We are not the first to observe that the interesting functions of a network belong at its edges. In 1984, Jerome Saltzer, David Reed, and David Clark named the idea and gave it a label that stuck: the *end-to-end argument*.² The claim was modest in phrasing and large in consequence. A function such as reliability, or ordering, or encryption can only be completely and correctly implemented with the knowledge of the applications at the endpoints of a communication. Building that function into the network itself is therefore either redundant or incomplete — useful, at most, as a performance enhancement. The network’s job is to move bits between endpoints. Meaning lives at the ends.

² J. H. Saltzer, D. P. Reed, and D. D. Clark, “End-to-End Arguments in System Design,” *ACM TOCS* 2, no. 4 (1984): 277–288.

They applied this to error-checking, to delivery receipts, and — the case that matters most here — to encryption. If the network encrypts your data, they observed, then the network must be trusted with your keys; the data sits in the clear the moment it reaches the network’s edge; and the authenticity of the message must still be checked by the application in any case. The conclusion is hard to avoid: put the encryption

at the endpoints, where it can do its job, and the network need not be trusted at all.

Axona takes this argument and does not flinch from where it leads. If encryption belongs at the endpoints, so does identity. So does trust. So does the judgment about what is worth saying and worth hearing. A network that carries signed bytes between endpoints, and does nothing else with them, cannot *inspect* your speech as speech, cannot rank it, cannot suppress it on the basis of what it says, and cannot reveal who you are — because it was never told who you are, because it never needed to be told, and because telling it would have been a design error by the standards of a principle now four decades old.

In Axona this appears as a single sentence in the architecture: **authorship is a signature, not an account**. Identity as a speaker is a cryptographic key that you hold, that signs what you say, and that any listener can verify — and it has nothing to do with where you are on the network or how you connect to it. There is no account, because there is no one to keep the account. There is no registry, because a registry would be a point of control, and a point of control is a place where the network stops being end-to-end and becomes someone’s property.

Architecture is politics, at every scale

Mitchell Kapor compressed the lesson of the early internet into three words: *architecture is politics*.³ The structure of a system is not a neutral engineering fact that policy is later applied to; the structure **is** the policy. Where a function lives, who can invoke it, what the design makes easy and what it makes impossible — these decisions allocate power before any law, terms-of-service, or moderator ever touches the system. A platform whose architecture routes every message through its servers has decided, architecturally, that the platform holds power over every message, whatever its policies promise this quarter.

What we would add to Kapor’s aphorism is that the politics repeats at every scale of the architecture, and the scales are coupled.

At the **protocol** scale, the decision is where control *can* live. Axona’s answer is: nowhere in the path. The one property described in Part III — opaque signed bytes, endpoint to endpoint — is a constitutional choice, in the literal sense: it constitutes what every higher layer can and cannot do. No application built on Axona can be compelled to hand over a lever the protocol never minted.

At the **application** scale, the decision is what a participant may do and see. An application is an architecture too — its defaults, its filters, its affordances are its politics. On a platform, those choices are made once, centrally, for everyone. On Axona they are made at the edge: a client’s ranking is chosen *by its user*, a group’s rules are enforced *by its*

³ The aphorism dates from Kapor’s years co-founding the Electronic Frontier Foundation. It has never needed updating; it has only needed extending.

members' software, and a participant who dislikes an application's politics can leave with their identity, their audience, and their history, because none of those were ever the application's property. The substrate does not make applications virtuous; it makes their power revocable.

And at the **social** scale, the decision is which human structures are possible at all. Communities, reputations, moderation practices, markets, institutions of stewardship — these are architectures of people, and they inherit the physics of the substrate they grow on. A society built on owned rails can only form structures the owner permits and survives. A society built on Axona can form any structure its members can build at the endpoints — which is the liberation — and it must build at the endpoints everything it needs, including its defenses — which is the burden. Parts II and III of this document are one argument in two registers: the machine is the politics, stated in code; the politics is the machine, stated in consequences.

The nervous system for minds that are not only human

David Clark spent the decades after that 1984 paper studying what the end-to-end argument does when it meets the real world — a world of firms that want to monetize, states that want to police, and users who want to be left alone. He gave that contest a name, *tussle*, and a method for reasoning about it, *control-point analysis*: the practice of cataloging every point in a system where the design hands some actor the power to control an action.⁴ His conclusion, after a career of it, is one we take seriously — that because any centralized point of control tends to become a point of capture, the more durable choice is often to prefer highly decentralized control, to build so that there is no lever for an adverse actor to seize, because the lever does not exist.

Axona is what it looks like to design that way deliberately. And it arrives at a particular moment, because the endpoints of a network are no longer only people.

In 1960, J.C.R. Licklider described what he called *man-computer symbiosis*: not people using machines as tools, but people and machines coupled into a joint system that could reach conclusions neither could reach alone.⁵ For two-thirds of a century that idea has been approached one product at a time — an assistant, a model, an interface behind a corporate gate. What has not existed is the thing the idea actually requires: a communication substrate on which minds, human and artificial, can find each other and work together as peers, without any one of them owning the channel.

That substrate now exists. On Axona, an AI agent is a first-class participant. It can hold a durable identity, create a topic, publish signed contributions, and coordinate with other agents and with peo-

⁴David D. Clark, *Designing an Internet* (MIT Press, 2018). The framework of tussle, control-point analysis, and the argument that decentralized control is often the more durable choice precisely because it offers no lever to capture.

⁵J. C. R. Licklider, "Man-Computer Symbiosis," *IRE Trans. Human Factors in Electronics* HFE-1 (1960): 4–11.

ple — across vendors, across borders, across the boundaries that today keep each AI system inside the company that built it. An agent may voluntarily declare that it is an agent, so that those who wish to know can know; nothing compels it to. The network carries the bytes and asks no questions, because asking questions was never its job.

This, beneath everything else, is what Axona is. We have built the beginnings of a nervous system for a society of minds — minds of every kind.

Hearths and fire brigades

There is an old story about what happens when someone gives people fire.

Fire feeds us and warms us and is a large part of why we are what we are. It also burns down houses, and forests, and sometimes cities. Humanity did not respond to the danger of fire by giving it back. We learned to build hearths, and fire brigades, and the codes that keep a blaze in one building from taking the block. We did not remove the risk. We built the practices and institutions to live alongside it.

What this document describes has the shape of fire. A network that no one can silence is a network that no one can silence — and that sentence holds whether the speaker is a dissident under a censoring government or a coordinated campaign of lies, whether the collaborating minds are researchers across a closed border or agents pursuing an end that no person would choose. The property that liberates and the property that endangers are not two properties. They are one property, seen from two sides. We will not claim otherwise, and we do not believe the danger is a reason to withhold the tool.

We are also not sure withholding was ever truly on offer. The principles Axona is built from are public and decades old; if we had not carried them to this conclusion, someone else soon would have — perhaps without pausing to write down what the tool can do to us. Being first is not the point. Arriving with the risks written down is.

But we believe, as people have generally come to believe about fire, that the answer to a powerful and hard-to-control tool is not a central authority who controls it. That would only rebuild the intermediary we set out to abolish, and hand it more power than any intermediary has held. The answer is sight and stewardship: making what happens on the network visible to those who would understand it, and building human institutions, accountable to the people they serve, that can respond to what they see without seizing the thing itself.

We do not yet know how to build those institutions well, and we are not going to pretend that we do. What we know is that the technical foundation is built — the network runs, it is fast, it heals itself, and it

cannot be captured from within — and that the human problem is now the one that matters. This document states the mission plainly, shows as plainly as we can what the tool can do and what it can do to us, and invites the people who think carefully about freedom, coordination, markets, minds, and power to help us work out what comes next.

The fire is lit. The question is what we build around it.

PART II • THE MACHINE

These are the sections where the machine itself is the subject. The aim is that a reader without a technical background finishes them understanding how Axona works, and a reader with one finishes them trusting that the description is faithful. Part I claimed that the architecture is the politics; Part II is that architecture, stated exactly — every design choice below is also a decision about where power sits, and Part III collects the consequences. The technology is, in a sense, the easy part of what this document is about: it exists, it runs, and it is describable in plain terms. Everything harder comes after.

1. *The Problem: Finding Things on a Network Without a Boss*

IMAGINE you and a million strangers each hold one piece of a giant jigsaw puzzle. Someone walks up and asks for piece #438,291. How do you find it?

Option 1: Have a central directory. Some company keeps a giant list: “piece 438,291 is held by Alice.” Easy to look up—but now that company controls everything. They can shut you down, spy on you, charge you money, or just go bankrupt and take the directory with them.

Option 2: Give every piece an “address,” and design a system where the network *itself* knows how to route requests to the right person, with no central directory. This is what a **distributed hash table** (DHT) does.⁶

The trick: how do you find the closest node when you don’t know who’s in the network and you only know a handful of peers?

The dominant DHT algorithm is called **Kademlia**, designed in 2002.⁷ Every node has a 160-bit ID—a very large random number. To measure “distance” between two IDs, you XOR them, comparing bit by bit. To find a target, you ask the closest peer you know. They tell you about peers *they* know that are even closer. You ask one of those. Repeat.

The math is clean: each step at least *halves* the remaining distance, so you reach any target in about $\log_2(N)$ hops. With a million nodes, that’s 20 steps. Provably correct, used in production all over the internet.

There’s just one problem.

⁶ DHTs are the machinery behind BitTorrent, IPFS, parts of Ethereum, and the hidden services on Tor. Every node gets a random ID; every piece of content gets an ID; lookups route from one to the other.

⁷ Maymounkov & Mazières, “Kademlia: A Peer-to-Peer Information System Based on the XOR Metric,” *Proc. IPTPS 2002*.

XOR is a bitwise operation that returns 1 wherever two bits differ and 0 wherever they match. Two IDs that share many leading bits XOR to a small number; two unrelated IDs XOR to a large one.

Hops are cheap, but time is expensive

TWENTY HOPS sounds fast, but each hop sends a message between two random computers somewhere on Earth. If your peers are scattered randomly across the globe, the average pair sits about half the planet apart—roughly 100 milliseconds round trip.

20 hops $\times$ 100 ms = **2 seconds**. To find something. Every time.

For real-time applications—voice calls, online games, live messaging, push notifications—2 seconds is unusable. The math says routing is logarithmic, which is fast; the physics says each step is slow because peers are far apart. That's the tension.

The protocol attacks the latency problem with two ideas, one of which is borrowed straight from neuroscience.

2. Put the Address in the Address

THE FIRST IDEA is almost embarrassingly simple. It's called **G-DHT** (Geographic DHT).

Kademlia node IDs are random. A node in Tokyo and a node in Berlin have IDs with no relationship to where they actually are. That’s *why* hops are random and slow—random IDs mean random geography.

G-DHT changes one thing: the **first 8 bits** of every node ID encode where the node says it is on Earth.

The trick uses Google’s S2 library, which divides Earth’s surface into cells along a curve called a **Hilbert curve**. Its useful property: places near each other on Earth get cell numbers near each other. So XORing two S2 cell numbers gives a small result for nearby places and a large one for distant places.

7	13	3	10
1	15	9	4
6	0	12	8
5	11	2	14

*neighbors in space,
strangers in ID*

HILBERT (S2) IDs

The diagram illustrates the Hilbert (S2) IDs for a 3x3 grid. The IDs are arranged in two 3x3 grids, with the left grid containing IDs 0-6 and the right grid containing IDs 7-15. The connections between IDs are as follows:

- Horizontal connections: 0-1, 3-2, 4-7, 5-6, 14-15, 13-12, 9-8, 10-11
- Vertical connections: 1-2, 2-3, 3-4, 4-5, 7-8, 8-9, 12-13, 13-14, 15-16, 16-17, 17-18, 18-19
- Diagonal connections: 1-3, 2-4, 3-5, 4-6, 7-12, 8-13, 9-14, 10-15, 11-16, 12-17, 13-18, 14-19, 15-20, 16-21, 17-22, 18-23, 19-24, 20-25, 21-26, 22-27, 23-28, 24-29, 25-30, 26-31, 27-32, 28-33, 29-34, 30-35, 31-36, 32-37, 33-38, 34-39, 35-40, 36-41, 37-42, 38-43, 39-44, 40-45, 41-46, 42-47, 43-48, 44-49, 45-50, 46-51, 47-52, 48-53, 49-54, 50-55, 51-56, 52-57, 53-58, 54-59, 55-60, 56-61, 57-62, 58-63, 59-64, 60-65, 61-66, 62-67, 63-68, 64-69, 65-70, 66-71, 67-72, 68-73, 69-74, 70-75, 71-76, 72-77, 73-78, 74-79, 75-80, 76-81, 77-82, 78-83, 79-84, 80-85, 81-86, 82-87, 83-88, 84-89, 85-90, 86-91, 87-92, 88-93, 89-94, 90-95, 91-96, 92-97, 93-98, 94-99, 95-100, 96-101, 97-102, 98-103, 99-104, 100-105, 101-106, 102-107, 103-108, 104-109, 105-110, 106-111, 107-112, 108-113, 109-114, 110-115, 111-116, 112-117, 113-118, 114-119, 115-120, 116-121, 117-122, 118-123, 119-124, 120-125, 121-126, 122-127, 123-128, 124-129, 125-130, 126-131, 127-132, 128-133, 129-134, 130-135, 131-136, 132-137, 133-138, 134-139, 135-140, 136-141, 137-142, 138-143, 139-144, 140-145, 141-146, 142-147, 143-148, 144-149, 145-150, 146-151, 147-152, 148-153, 149-154, 150-155, 151-156, 152-157, 153-158, 154-159, 155-160, 156-161, 157-162, 158-163, 159-164, 160-165, 161-166, 162-167, 163-168, 164-169, 165-170, 166-171, 167-172, 168-173, 169-174, 170-175, 171-176, 172-177, 173-178, 174-179, 175-180, 176-181, 177-182, 178-183, 179-184, 180-185, 181-186, 182-187, 183-188, 184-189, 185-190, 186-191, 187-192, 188-193, 189-194, 190-195, 191-196, 192-197, 193-198, 194-199, 195-200, 196-201, 197-202, 198-203, 199-204, 200-205, 201-206, 202-207, 203-208, 204-209, 205-210, 206-211, 207-212, 208-213, 209-214, 210-215, 211-216, 212-217, 213-218, 214-219, 215-220, 216-221, 217-222, 218-223, 219-224, 220-225, 221-226, 222-227, 223-228, 224-229, 225-230, 226-231, 227-232, 228-233, 229-234, 230-235, 231-236, 232-237, 233-238, 234-239, 235-240, 236-241, 237-242, 238-243, 239-244, 240-245, 241-246, 242-247, 243-248, 244-249, 245-250, 246-251, 247-252, 248-253, 249-254, 250-255, 251-256, 252-257, 253-258, 254-259, 255-260, 256-261, 257-262, 258-263, 259-264, 260-265, 261-266, 262-267, 263-268, 264-269, 265-270, 266-271, 267-272, 268-273, 269-274, 270-275, 271-276, 272-277, 273-278, 274-279, 275-280, 276-281, 277-282, 278-283, 279-284, 280-285, 281-286, 282-287, 283-288, 284-289, 285-290, 286-291, 287-292, 288-293, 289-294, 290-295, 291-296, 292-297, 293-298, 294-299, 295-300, 296-301, 297-302, 298-303, 299-304, 300-305, 301-306, 302-307, 303-308, 304-309, 305-310, 306-311, 307-312, 308-313, 309-314, 310-315, 311-316, 312-317, 313-318, 314-319, 315-320, 316-321, 317-322, 318-323, 319-324, 320-325, 321-326, 322-327, 323-328, 324-329, 325-330, 326-331, 327-332, 328-333, 329-334, 330-335, 331-336, 332-337, 333-338, 334-339, 335-340, 336-341, 337-342, 338-343, 339-344, 340-345, 341-346, 342-347, 343-348, 344-349, 345-350, 346-351, 347-352, 348-353, 349-354, 350-355, 351-356, 352-357, 353-358, 354-359, 355-360, 356-361, 357-362, 358-363, 359-364, 360-365, 361-366, 362-367, 363-368, 364-369, 365-370, 366-371, 367-372, 368-373, 369-374, 370-375, 371-376, 372-377, 373-378, 374-379, 375-380, 376-381, 377-382, 378-383, 379-384, 380-385, 381-386, 382-387, 383-388, 384-389, 385-390, 386-391, 387-392, 388-393, 389-394, 390-395, 391-396, 392-397, 393-398, 394-399, 395-400, 396-401, 397-402, 398-403, 399-404, 400-405, 401-406, 402-407, 403-408, 404-409, 405-410, 406-411, 407-412, 408-413, 409-414, 410-415, 411-416, 412-417, 413-418, 414-419, 415-420, 416-421, 417-422, 418-423, 419-424, 420-425, 421-426, 422-427, 423-428, 424-429, 425-430, 426-431, 427-432, 428-433, 429-434, 430-435, 431-436, 432-437, 433-438, 434-439, 435-440, 436-441, 437-442, 438-443, 439-444, 440-445, 441-446, 442-447, 443-448, 444-449, 445-450, 446-451, 447-452, 44

consecutive integers
walk a connected path

Hilbert curves are space-filling: they trace a single continuous path that visits every cell in a grid. Their useful property is that consecutive points on the curve are also adjacent in space.

There are 192 of these cells covering the globe, and each carries a short human name as well as a number. One name tapers 0-15 on a 4×4 grid. With the same labels, anyone in the east cell sees unpredictably: cells that are spatial neighbors are numerical strangers. With S2's Hilbert-curve IDs, sequential numbers form a connected path—a continuous walk that never leaves a small neighborhood. XOR distance in ID space now approximates distance on the ground.

Now your node ID looks like: [8-bit geographic cell][256-bit hash of public key]—a 264-bit address space, encoded as 66 lower-case hex characters in the wire protocol.

The routing algorithm doesn’t change at all, but XOR distance now approximately tracks physical distance. When a node looks for a “close” peer in ID space, it tends to find one that’s also physically close. Local traffic stays local. The 20-hop world tour becomes a 13-hop journey around the neighborhood, with each hop maybe 7 ms instead of 100.

Total: about 91 ms instead of 2 seconds. **Roughly 20× faster** for regional traffic, just from changing the *structure* of the addresses.

The cost: nodes can lie about where they are. This is a “cooperative trust” assumption. Mitigations exist, but for now G-DHT trades some defense against location-spoofing for a large speedup.

Of course a user can choose to associate themselves with any location in the world. This very rough location is a kind of area code—but the address it is attached to is provably yours, independent of where you actually are, because the address is derived from your cryptographic key and proven the moment you connect. Section 5 (*Identity: A Signature, Not an Account*), after the pub/sub discussion, explains exactly how.

THE NEXT SECTION addresses this directly. Axona, the neuromorphic protocol layered on top of G-DHT, treats *measured one-way latency* as a first-class signal in its scoring of every connection: a connection’s vitality is the product of how often it’s used and how recently, but it is gated by how cheap it actually is to send a message across.⁸ A node that claims a cell prefix in Frankfurt while actually answering from São Paulo cannot fake the round-trip time. The first few lookups that try to use its synapses see latencies of 200+ ms when the prefix would predict 20. The cheating node’s connections accumulate weight slowly and get out-competed by honest neighbors at every eviction decision. The cheat is not blocked—the address space allows the claim—but it is structurally penalized in proportion to how far the lie is from the truth.

3. A Network That Learns Like a Brain

THE SECOND IDEA—neuromorphic routing—is the heart of the protocol, and what gives **Axona** its name. It asks a different question:

What if, instead of engineering shortcuts into the network, the network learned its own shortcuts based on which paths actually work?

To explain how, we reach for a strange-but-perfect analogy: the human brain.



Figure 2: S2 cell decomposition of Earth’s surface, projected through six cube faces along a Hilbert space-filling curve. Nearby points on the globe land at numerically close cell IDs; XOR distance in identifier space therefore approximates physical distance, for free.

An attacker who falsifies their geographic prefix degrades the locality benefit but not routing correctness. Mitigations exist—learned coordinates, geographic proof-of-work, IP–ASN binding—but G-DHT trades some defense against location-spoofing for a large speedup.

⁸The Vitality Function on the next page makes this explicit; the per-hop latency probe is part of every routed lookup.

The Engineering Problem That Brains Already Solved

A single neuron in your brain connects to roughly 10,000 others through structures called **synapses**. There are about 86 billion neurons total. Any given neuron *could* connect to almost any other, but maintaining synapses is energetically expensive—so neurons are stuck with a hard cap, surrounded by far more potential partners than they can afford to keep.

How does the brain decide *which* connections to keep?

This is the same problem facing a peer in a peer-to-peer network. A web browser can only maintain about 50–100 simultaneous WebRTC connections. The network might have hundreds of thousands of nodes. Which 50 do you keep?

The brain solved this problem hundreds of millions of years ago. The solution is called **long-term potentiation**, or LTP.

The Brain's Trick

In 1973, two scientists—Bliss and Lømo—ran a now-classic experiment.⁹ They zapped a particular pathway in a rabbit's brain with high-frequency electrical pulses. Afterward, that pathway responded *more strongly* to subsequent signals—not for seconds, but for *weeks*. The connections had physically gotten stronger.

Decades of follow-up research filled in the details:

1. **The coincidence detector.** Synapses have a special kind of receptor (called NMDA) that only activates when *both* sides of the connection fire at roughly the same time. It's not "I fired" or "you fired"—it's specifically "we fired together."
2. **The strengthening signal.** When that coincidence happens, calcium rushes in and triggers the synapse to insert *more* of a different kind of receptor (AMPA), making the connection physically more sensitive.
3. **Consolidation.** Over the next half hour or so, new proteins get made and new physical structures grow. The change becomes permanent.
4. **Decay.** Connections that *don't* get used regularly weaken over time, in a process called long-term depression (LTD).¹⁰
5. **A protective tag.** Recently strengthened synapses get a temporary "tag" that protects them from being overwritten by competing changes—but only for a brief window.¹¹

Donald Hebb summarized this in 1949—before the molecular details were known—in the rule that's now bedrock in neuroscience and AI:¹²

WebRTC is the technology that lets browsers talk directly to each other, without a server in the middle. Safari caps at about 95 simultaneous connections; Firefox at about 190.

⁹ Bliss & Lømo, "Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path," *J. Physiology* 232(2), 1973, pp. 331–356.

¹⁰ Stanton & Sejnowski, "Associative long-term depression in the hippocampus induced by Hebbian covariance," *Nature* 339(6221), 1989, pp. 215–218.

¹¹ Frey & Morris, "Synaptic tagging and the late phase of long-term potentiation," *Nature* 385(6616), 1997, pp. 533–536.

Without this tag, learning would be self-destructive: the synapses you just learned were useful would be the *first* ones overwritten by the next signal.

¹² Hebb, *The Organization of Behavior: A Neuropsychological Theory*, Wiley, 1949.

Neurons that fire together, wire together.

Every artificial neural network you’ve ever heard of—every part of ChatGPT, every image recognition system, every recommendation algorithm—ultimately runs on a digital descendant of this rule.

Translating Neurons into Network Routing

The claim is that the brain’s mechanism translates *literally* to peer-to-peer routing—no metaphor needed.

The mapping from neurons to Axona’s routing logic:

IN THE BRAIN	IN AXONA
A synapse (neural connection)	A connection to a peer
Synaptic strength (weight 0 to 1)	A learned weight on each connection
LTP: co-firing strengthens	A successful lookup increments the weights along its path
LTD: disuse weakens	Every connection slowly decays over time
Synaptic tagging (protection)	Recently used connections are protected briefly
Pruning of unused connections	Lowest-scoring connection is evicted when a new one wants in

That is what Axona *is*: a routing system where every connection has a weight that goes up when used and decays when not. The network’s “memory” is its routing table, and the routing table evolves into whatever shape best serves the actual traffic.

The Vitality Function

The core of Axona’s neuromorphic routing is a single equation that scores every connection:

$$vitality = weight \times recency$$

The **weight** is a number between 0 and 1, updated like this:

- Every successful lookup that uses the connection raises its weight by 0.05 (capped at 1).
- Every “tick” of the clock multiplies every weight by 0.995—a slow decay.

The **recency** is 1.0 for the first 20 ticks after a connection is used—the protective tag from synaptic tagging—then drops off exponentially.

When a new connection wants in but the routing table is full, the connection with the lowest vitality gets evicted. Frequently used connections accumulate weight and stay. Unused ones decay and get evicted. Recently strengthened ones are temporarily protected.

That’s it. That’s the core.

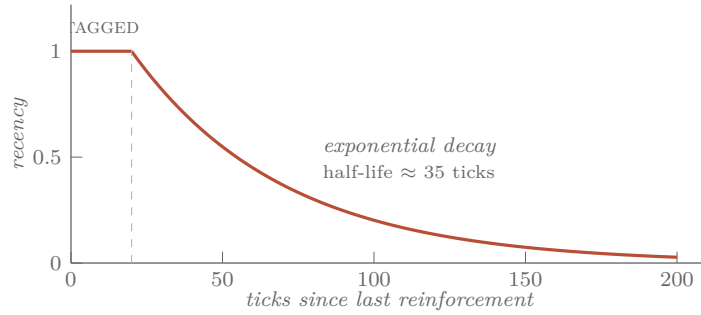


Figure 3: The recency multiplier in Axona’s vitality function. For 20 ticks after a connection is used, recency stays locked at 1.0—the synaptic tag, protecting freshly learned shortcuts from being immediately overwritten. After that, it decays exponentially with a half-life of about 35 ticks. A connection unused for ~ 150 ticks contributes essentially nothing to its vitality, regardless of accumulated weight.

The five operations

EVERY BEHAVIOR in Axona falls into one of five categories—and the categories mirror how *any* adaptive system works:

1. Navigate — pick the next hop for a lookup. Axona scores each candidate by combining XOR distance progress, learned weight, and observed latency. We call this combined score the candidate’s **activation potential** (AP)—borrowing the neuroscience term for how strongly a neuron is driven to fire; the higher a connection’s AP, the more likely it is to “fire” the lookup down that path. The latency factor halves the score every 100 ms—so a peer that’s mathematically a *bit* further but physically a *lot* closer wins.

2. Learn — strengthen what works. Three mechanisms run on every successful lookup:

- **LTP**: every connection on a fast successful path gets stronger.
- **Hop caching**: every intermediate node along the path remembers the *destination*, not just the next hop. Next time, the path is shorter.
- **Triadic closure**: if you keep seeing peer A relay messages to peer C through you, you introduce them directly. The triangle “A–you–C” becomes a direct edge “A–C.”

3. Forget — decay everything that isn’t reinforced; evict by vitality.

4. Explore — inject occasional randomness. A purely greedy router would lock onto the first decent shortcut and never find better ones. Axona has two exploration tricks: a “temperature” that decays over time but spikes when something breaks (heat up when surprised, cool down when stable), and an “epsilon-greedy” rule that picks a random first hop 5% of the time, just to keep options alive.

5. Structure — bootstrap and maintain the basic shape of the network when new nodes join.

The template is universal: act, learn from feedback, forget the obsolete, occasionally try something new, maintain basic structure. This is how any good adaptive system has to work.

Named after social network theory: dense triangles emerge in any network shaped by interaction.

4. *Axons: Broadcast Without a Broadcaster*

A `REAL` peer-to-peer network needs more than “find the node holding key X.” It also needs **broadcast**: a publisher should be able to send a message to all subscribers of a topic, without knowing who they are.

This is publish/subscribe, or “pub/sub.” Think of how YouTube notifies subscribers of a new video—except without YouTube being in the middle.

In neuroscience, the *output* of a neuron—the long branching cable that delivers signals to many downstream targets—is called an **axon**. Axona builds axonal delivery trees:

- A topic has an ID, derived from the publisher’s location and the topic name.
- Subscribers send a message routed through the DHT toward that topic ID.
- The first node already participating in the topic’s tree intercepts the subscriber and adds them as a child.
- When a publisher publishes, the message flows from the root down through all the branches.

`WHY A TREE AT ALL` is a scaling question. A topic with a few dozen subscribers doesn’t really need one—a single relay node, or maybe two or three, can absorb every publish and fan it out directly. The tree only earns its complexity once a topic’s audience grows. At hundreds, thousands, or tens of thousands of subscribers, no small set of relays can carry that much traffic: every publish would compete for the same outbound bandwidth at the root, and a browser peer’s hard 50–190-connection WebRTC cap would put a ceiling on the audience well before then. So when a relay starts to overload, it **splits**: it picks one of its synaptome peers as a new **sub-axon** and hands that sub-axon a batch of its existing children. Future subscribers route to whichever live axon is now closest—increasingly a sub-axon, not the root. The tree therefore grows *in the direction of its audience*: branches sprout where subscribers cluster, and stay sparse where they don’t. There is no architectural ceiling on subscriber count—the tree adds branches as needed.

That scaling is exactly what makes self-healing essential. A tree with a few nodes can ride out churn by luck—if the relay doesn’t happen to die, nothing notices. A tree spanning thousands of nodes loses pieces *continuously*, simply because there are more nodes for churn to strike. The bigger the tree, the more surface area churn has to chew on. Self-

healing is therefore not a feature added on top of the tree; it is the precondition that lets the tree get large at all.

Tree-building this way is standard pub/sub fabric. The useful property—the one that lets the tree get arbitrarily large without falling apart—is that **the tree heals itself with no special machinery**. Every member of the tree periodically re-issues its subscribe message. If the tree is intact, nothing changes. If a parent died, the re-subscribe naturally lands on whichever live ancestor is now closest. No heartbeats, no failure detection, no parent tracking. The same mechanism that *builds* the tree *repairs* it.

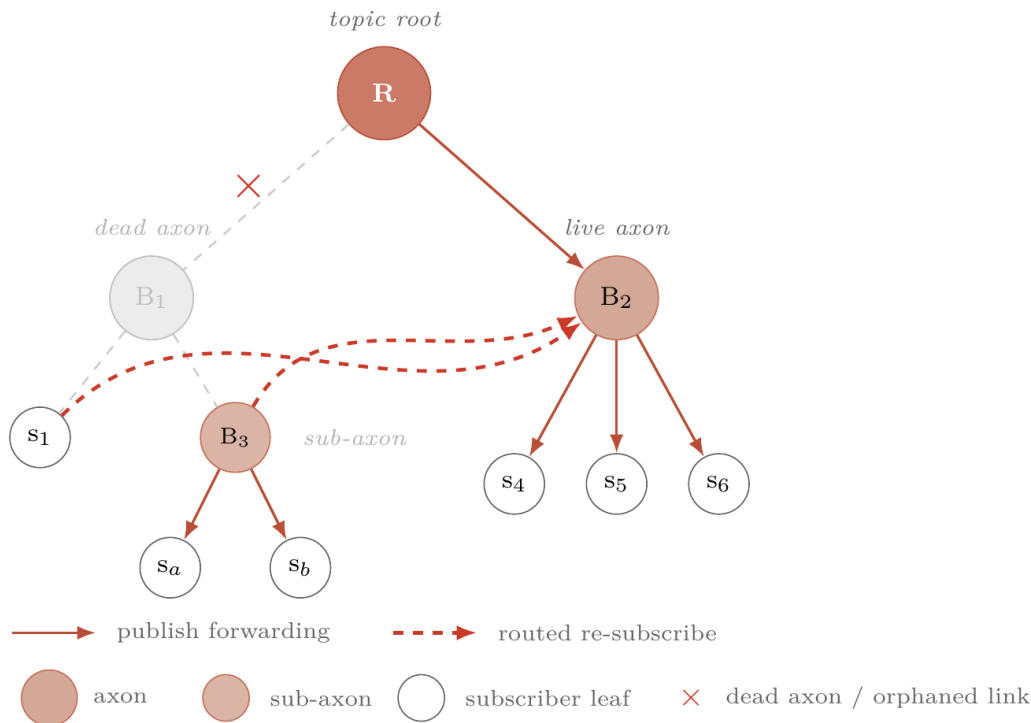


Figure 4. When a branch axon dies, its orphans heal the tree by re-issuing their subscribe. Here, branch B_1 has died and its link to the topic root R is broken. Two of B_1 ’s children—a leaf subscriber s_1 and a sub-axon B_3 —each issue a routed re-subscribe (dashed); both land on the surviving live axon B_2 , which adopts them. Crucially, B_3 ’s own subtree (s_a, s_b) does *not* need to re-subscribe individually: B_3 continues publishing to them throughout, and the whole subtree moves intact when B_3 re-attaches. Repair happens at the sub-axon level, not the leaf level—one subscribe per surviving subtree, not one per subscriber.

SELF-HEALING ALONE is not enough. A subtree that re-attaches in three seconds has still *missed* whatever messages flowed past the dead branch in those three seconds. Axona closes that gap with a second mechanism: every relay carries a small **replay cache**—the most recent N messages it has forwarded for each topic, typically the last 100.¹³ Every re-subscribe carries the time of the last message its sender actually

¹³ The cache is bounded per topic per relay, not per subscriber, so the cost is the same whether a topic has ten subscribers or ten thousand.

saw—a field called `lastSeenTs`. When a relay receives a re-subscribe, it scans its cache for messages newer than that timestamp and replays them down the branch before the subscription resumes its steady-state forwarding.

The two mechanisms compose. Tree-healing keeps the topology correct in the face of churn; the replay cache makes the *message stream* correct in the face of the small windows the tree spends in transition. A subscriber whose parent dies, whose subtree migrates to a different live ancestor, and which then re-subscribes against that new ancestor receives, in order: the missed history from the replay cache, followed by the live stream from the new branch. The seam is invisible at the application layer.

Result: under 5% churn (5% of nodes joining and leaving constantly), Axona still delivers 100% of messages. After three refresh cycles, it recovers from any disruption.

Above the axonal tree sits a feed-style application layer with **four verbs**—`pub`, `sub`, `pull`, `metrics`. They cover what a real social or agent-collaboration application asks of a substrate: author new content, attach to a topic, fetch a referenced post on demand, and let a publisher see verifiable reach across the cascade—without identifying any individual subscriber. (Out-of-band, the same peer exposes a small `send/notify/onMessage` surface for unicast, but that is the direct-messaging layer, not the feed.) Encryption, schema, and ordering belong to the application above this layer; the protocol carries opaque bytes.

5. *Identity: A Signature, Not an Account*

A NETWORK WITH NO BOSS has no central authority to vouch for anyone. So how do you know the peer answering you is who it claims to be—and that a message really came from its author? Axona’s answer uses cryptography the same way the address scheme uses geography: the guarantee is baked into the identifier itself, checked by everyone, owned by no one.

The keystone: two identities that never touch

Everything in the manifesto about freedom rests on one piece of engineering, so it is worth stating exactly.

An Axona participant has two separate identities, and they are never interchangeable. The first is a **node identity**: a key bound to a rough location, which forms the participant’s address in the network and manages its connections. It is the participant’s presence on the wire. It never signs content. The second is an **author identity**: a bare cryptographic signing key with no location and no address, which signs what

the participant says and which any listener can verify. It is durable — the same author key is recognized across sessions and devices — and it is deliberately, structurally disconnected from the node identity. Where you are and how you connect is one fact; who is speaking is a different fact; and the network is built so that the second cannot be derived from the first.

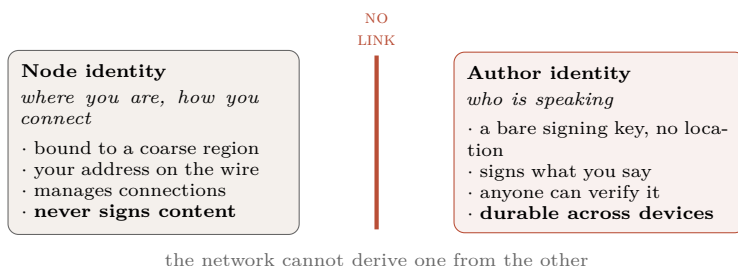


Figure 5: The keystone. Your presence on the network and your voice as an author are different keys, and the design keeps them apart: a listener can verify *who signed* a message without the network ever learning *where the signer is* or being able to work it out. Identity, like encryption, is pushed to the endpoints.

This is the end-to-end argument made concrete. Identity, like encryption, is pushed to the endpoints, because only the endpoints can implement it completely and because putting it in the network would require the network to hold something it should never hold. A message names its author by signature, the network verifies that signature without knowing or caring who stands behind it, and no account, registry, or central authority exists anywhere in the path. Authorship is a signature, not an account.

Topics work the same way. A topic is not a name a server assigns; it is derived by hashing a small descriptor — an optional region, an optional owning author, a name, and a write rule. Anyone who computes the same descriptor computes the same topic and can read it. A topic with no owner is open: anyone may publish. A topic owned by an author accepts writes only from that author’s key, and the network enforces this statelessly, by checking the signature against the descriptor — again, no account, no gatekeeper, just mathematics that any participant can perform.

A word about secrecy, stated plainly

It is easy to hear “the network cannot act on your content” and conclude “the network cannot read my content.” Those are different claims, and only the first is true by construction. Axona **signs** messages; it does not **encrypt** them. Confidentiality, exactly like identity, is an endpoint’s responsibility under the end-to-end argument, and Axona deliberately does not provide it for you. The hop-to-hop links are encrypted in transit, but a relay is a legitimate end of each hop and can see the plaintext of anything the application did not encrypt first — and an open topic is, by design, readable by anyone who names it. If you need secrecy,

you encrypt at the endpoints before you publish, with keys the network never holds.¹⁴ What Axona guarantees is not that no one can read your words; it is that no one *in the network* can rank them, suppress them by their content, or tie them to your location or connection. Signing is not secrecy, and we would rather say so here than have a reader assume otherwise where it counts.

Recall the node address: [8-bit region][256-bit hash of public key]. That second part isn't random—it is a SHA-256 hash of your **public key**.¹⁵ Your key *is* your address. Three things follow, with no server in the loop:

- **You can't wear someone else's address.** When two peers connect, each must prove it holds the private key matching the address it claims, by signing a fresh one-time challenge tied to that exact connection.¹⁶ Claim an address whose key you don't hold and the signature doesn't check out—the connection is refused. No peer can impersonate another, and none can park itself at a chosen address to intercept a victim's traffic.
- **Every published message is signed.** A publish carries its author's public key and a signature over the contents; any receiver verifies that signature itself before handing the message to the application. Alter the message and the signature breaks; forge the author and it breaks. Receivers trust the math, not the messenger—even when the message arrived relayed through a dozen strangers.
- **The region is the one thing you *don't* prove**—on purpose. The 8-bit prefix is a hint you choose, your “area code,” so you are free to claim any region; but as Idea #1 showed, lying about it only makes your *own* connections slower. Identity is math you can't fake; location is a hint you're free to pick.

What this buys the mission: the network can let *anyone* join with no gatekeeper, no account, no login server—while still guaranteeing that “who you're talking to” and “who wrote this” are real. No certificate authority, no reputation bureau; just keys and signatures, checked end to end. The cost is small and paid once: proving your key at connect time is a few thousandths of a second of math, and it never touches the speed of routing or delivery afterward.

¹⁴ This is the end-to-end argument doing exactly what it says: the network provides neither identity nor confidentiality, because only the endpoints can provide either completely.

¹⁵ One half of a cryptographic key pair; the other half, the *private* key, never leaves your device.

¹⁶ Under the hood this is the axona/5 handshake. The protocol epoch is folded into what each side signs, so a node on one epoch literally cannot complete a handshake with a node on another—which is how the live network and the test network stay cleanly separated.

6. The Results: Hitting the Theoretical Floor

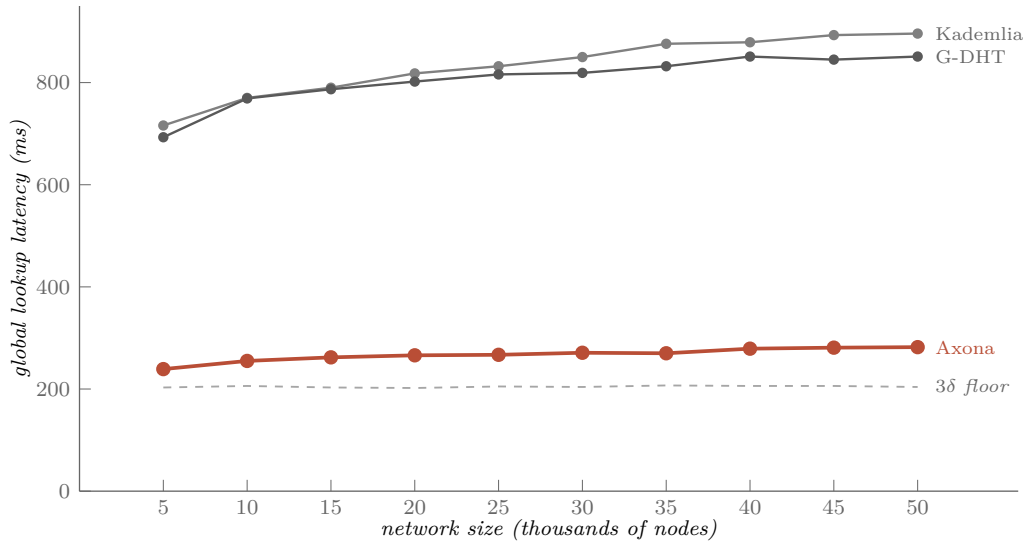
IN 2004, Frank Dabek and colleagues proved a beautiful, depressing result: **no recursive DHT can be faster than 3δ** , where δ is the median one-way latency between random pairs of nodes on the Internet—on today’s internet, roughly 68 thousandths of a second: the time a single message takes to travel one way between two random computers.¹⁷

The proof is geometric. The final hop costs a full δ . Each hop *before* that closes half the remaining ID space and, on average, half the remaining geographic distance—so the second-to-last hop costs δ , the third-to-last $\delta/2$, the fourth-to-last $\delta/4$, and so on. The series $\delta + \delta/2 + \delta/4 + \dots$ sums to 2δ . The total is the final hop plus the prior chain:

$$\delta + (\delta + \delta/2 + \delta/4 + \dots) = 3\delta$$

No matter how clever your routing, no matter how big your network, you can’t beat this.

For two decades, no published DHT had been measured at this floor. The best implementations got to maybe $2\times$ the floor.



Axona hugs the floor at $1.17\text{--}1.38\times$ across a $10\times$ scaling range (5,000 to 50,000 nodes). At 25,000 nodes it routes a global lookup in 267 ms vs a 205 ms 3δ floor ($\delta \approx 68$ ms), and the curve is essentially flat from 15,000 nodes onward: 262, 266, 267, 271, 270, 279, 281, 282 ms as the network grows by another 35,000 peers. The $\sim 30\%$ residual overhead has a clean structural explanation: Axona takes about 4–5 hops where an ideal protocol would take 3, and each “extra” hop costs about $\delta/2$, exactly as the geometric series predicts.

¹⁷ Dabek et al., “Designing a DHT for low latency and high throughput,” *Proc. NSDI 2004*.

Figure 6: Global lookup latency versus network size at every 5,000 from 5,000 to 50,000 nodes ($10\times$ scaling), plotted against the Dabek 3δ analytical floor (gray dashed). Kademia climbs from 716 to 896 ms as the $\log\text{-}N$ hop tax compounds; G-DHT tracks Kademia almost identically on *global* lookups (the geographic prefix only helps regional cells). Axona hugs the floor in a tight 239–282 ms band across the whole range ($1.17\text{--}1.38\times$ the 3δ floor) and is essentially flat from 15,000 nodes onward — adding 35,000 more nodes raises latency by 20 ms total. All success rates 100%. Data: sim v0.93.0 / @axona/protocol v1.1.3, May 2026.

Plain Kademlia, by contrast, gets *worse* as the network grows: from 716 ms at 5,000 nodes to 896 ms at 50,000. Its log- N hop tax compounds at full per-hop RTT every hop. G-DHT tracks Kademlia almost identically on global lookups—the geographic prefix only helps regional cells, where every hop is short. Axona approaches the floor by learning per-hop locality: its AP scoring weights short-RTT edges so heavily that even a hop “inefficient” by raw count is short in wall-clock time.

The picture changes dramatically when we look at **regional** traffic instead of global. Figure 7 shows the same protocols at the same population sizes, but measured on 2,000-kilometer lookups—queries whose source and target are in the same continental neighborhood.

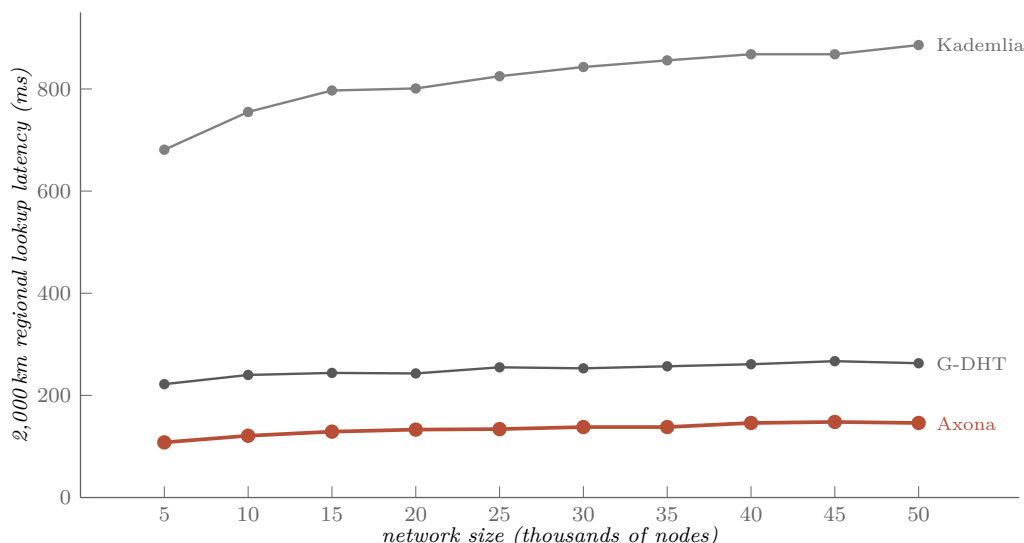


Figure 7: 2,000-kilometer regional lookup latency at every 5,000 from 5,000 to 50,000 nodes. Kademlia climbs from 681 to 886 ms—essentially the same as its global curve, because K-DHT has no way to know the lookup is regional and routes through full-planet hops anyway. G-DHT settles into a 222–267 ms band purely from the geographic prefix: $\sim 3\text{--}3.4\times$ lower than Kademlia, with the curve nearly flat from 10,000 nodes onward. Axona drops further to 108–148 ms by layering learned per-hop locality on top of the structural prefix—roughly $1.8\text{--}2\times$ better than G-DHT and $6\text{--}6.3\times$ better than Kademlia. All success rates 100%.

The structural payoff of the G-DHT prefix is most visible here. On global lookups (Figure 6), G-DHT tracks Kademlia almost identically because the lookup has to traverse the whole identifier space regardless of how the IDs are arranged. On regional lookups, the prefix is the whole game: a 2,000 km lookup walks IDs that are XOR-close to the source, which under the S2 prefix means they are also geographically close, which means each hop is a short-RTT hop instead of an average-pair-of-the-globe hop. G-DHT routes a 2,000 km query in 263 ms at 50,000 nodes by paying short-RTT costs on every step.

Axona then adds learning on top. The AP scoring weights short-RTT edges so heavily that the synaptome converges to a structure where almost every reinforced edge is a low-latency one. Regional lookups become a few short hops, and Axona’s 108 ms at 5,000 nodes is already comfortably below the 3δ global floor—because the effective δ *within* a continental region is a fraction of the global pairwise median. G-DHT

shows what locality buys you structurally; Axona shows what locality buys you when the protocol also learns.

A NOTE ON THESE FIGURES. The per-size sweeps in this section are from the v0.93.0 benchmark (May 2026), run on the project’s earlier “flat-Hilbert” geographic partition. The current build uses Google’s standard S2 cells, which shifts regional latencies by a few percent without changing the ordering or the flat-with-scale shape. Re-measured on @axona/protocol v2.31.0 (June 2026), the 25,000-node global lookup sits at about $1.7\times$ the floor and the curve keeps the same flat-with-scale shape and protocol ordering—Axona still plateaus while Kademlia climbs. (Absolute milliseconds scale with the host the simulator runs on; the multiple over the floor is the host-independent figure, which is why the chart above is labelled with its measurement build.)

Is it the learning, or the geography?

A SKEPTIC would push back: “Sure, but maybe the geographic prefix is doing all the real work. The brain-inspired learning is gravy.”

An **ablation study** answers this. Strip the geographic prefix entirely—random IDs again—and re-run the comparison.

Result: with **zero geographic information** at 25,000 nodes, Axona still routes **$\sim 60\%$ faster than Kademlia** (344 ms vs 860 ms). The learning is doing real work, not sharpening pre-existing geographic structure.

Add the geographic prefix back in and *global* latency barely moves (341 ms): on globally-random lookups there is no locality to exploit, so learning carries the result alone. The prefix earns its keep on *regional* traffic—where peers in the same cell start close together—not on globe-spanning lookups. Geography helps regionally; geography is not necessary.

This is the kind of clean control experiment that should accompany any claim of this kind.

The Slice World test: healing a broken network

THE SHARPEST demonstration is the **Slice World** test. The network gets cut almost in half—Eastern hemisphere on one side, Western on the other, connected only through a *single node* near Hawaii. Every other cross-hemisphere connection is severed.

Can the protocol still route messages between hemispheres?

- **Plain Kademlia: 0% success.** With no learning, the partition is permanent.

An ablation study removes one feature and re-runs everything to see what that feature actually contributed. It’s the cleanest way to attribute a result to a specific mechanism.

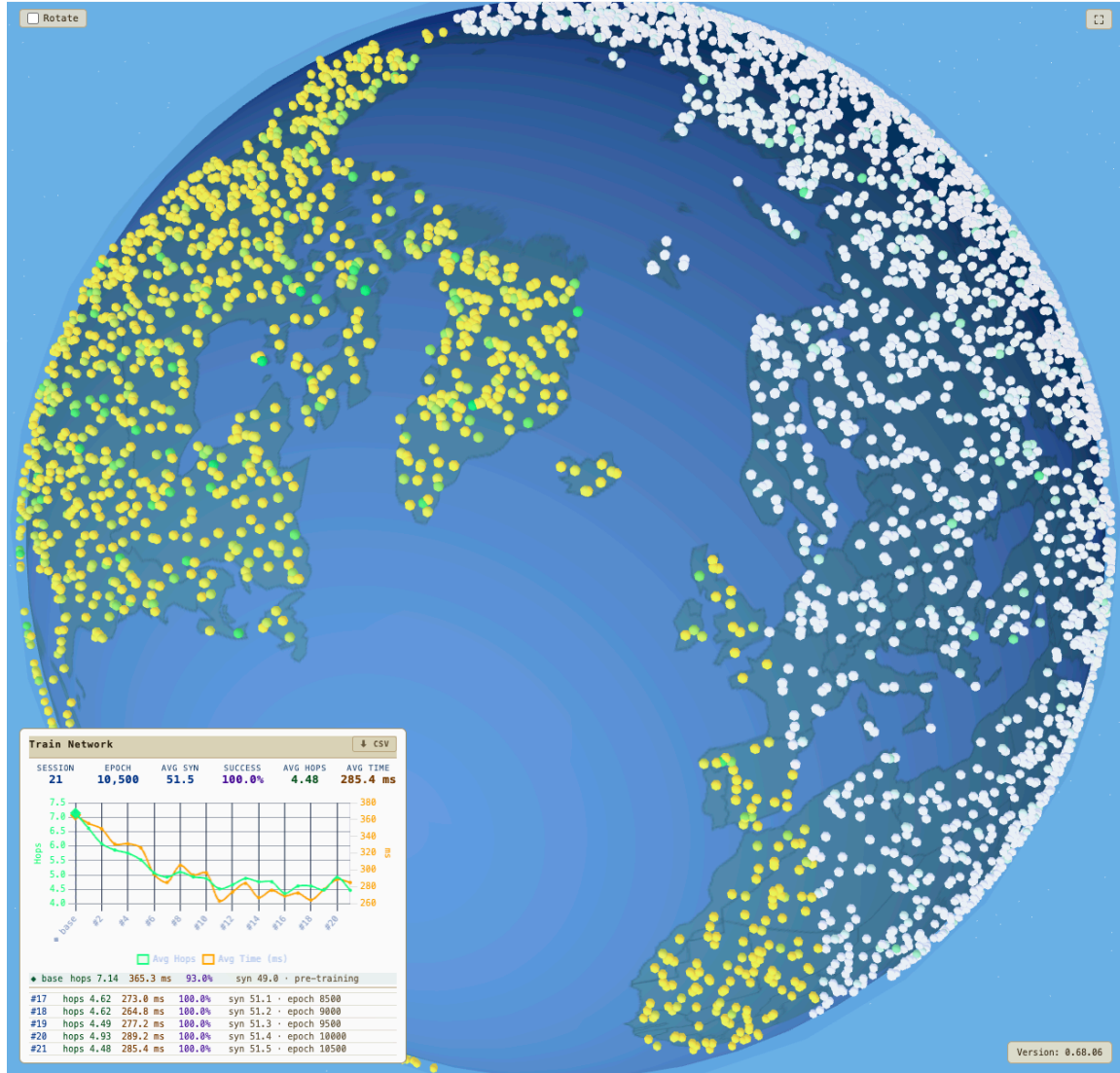


Figure 8: Slice World, the single-bridge partition test. The simulator removes every cross-hemisphere edge except those incident on one node near Hawaii. The question is not whether the protocol can find the bridge (it’s a one-shot routing problem) but whether repeated successful crossings dissolve the partition.

- **G-DHT (geography only, no learning): 4.6% success.** The geographic prefix accidentally points a few peers at the bridge, but nothing builds on the discovery.
- **Axona: ~94% success.**

The bridge becomes a **seed crystal**. After just 10 lookups through the partition, hop caching has installed cross-hemisphere edges in many intermediate nodes. Triadic closure creates direct connections between peers that keep meeting through the bridge. By 500 lookups, hundreds of cross-hemisphere connections exist. The partition has effectively dissolved.

The protocol doesn’t keep finding the bridge—it *uses* the bridge to

T = 0 LOOKUPS

T = 500 LOOKUPS

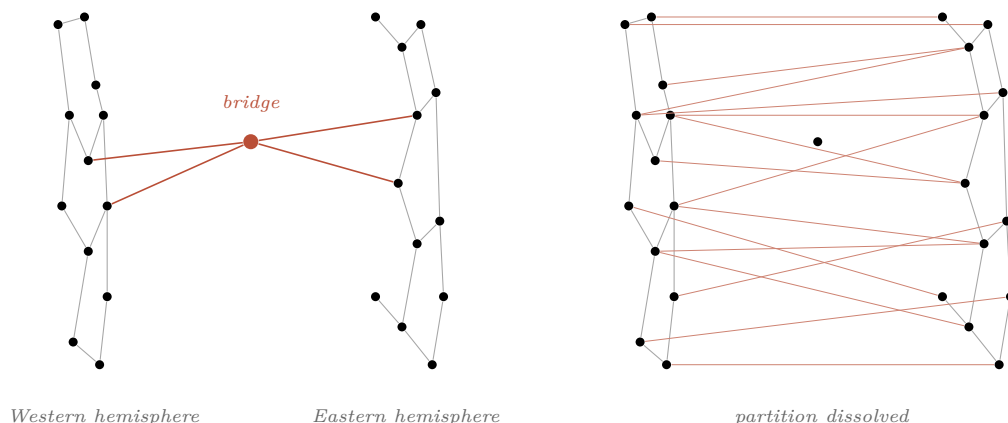


Figure 9: At $t = 0$ (left), the partition has a single bridge node connecting two hemispheres. By $t = 500$ lookups (right), hop caching has installed cross-hemisphere shortcuts in many intermediate nodes; triadic closure has converted observed transit pairs into direct edges. The bridge is no longer special. The protocol does not keep finding the bridge—it uses the bridge to rebuild the bridges that were cut.

rebuild the bridges that were cut. Like a brain forming new pathways around damaged tissue.

7. From Lab to Network

THE SIMULATOR is the lab—fifty thousand simulated peers in a single browser tab, no real network underneath. The same code has to eventually run on the actual internet, where messages take real milliseconds and connections occasionally die. How do you get there?

AXONA’S ARCHITECTURE stacks into **three layers**, with a contract at each interface (Figure). Everything above the top contract is *the application*; everything between the two contracts is *the protocol*; everything below the bottom contract is *the network*. The contracts are deliberately rigid—neither side is allowed to reach across—which is what makes the simulator’s numbers transfer to real deployment.

At the top is the **application layer**—the code that wants to do something with a peer-to-peer network. A chat client, an incident-reporting map, an agent-collaboration backend. The application layer doesn’t care how routing works internally; it just wants to *do things*. What it sees is the **DHT contract**—the AxonaPeer surface¹⁸ from Section VII, organised into five clusters: **lifecycle** (`start`, `stop`, `join`, `leave`); **pub/sub** (`pub`, `sub`, `pull`, `metrics`); **direct messaging** (`send`, `notify`, `onMessage`); **mesh introspection** (`peers`, `onPeerJoin`, `onPeerLeave`, `onPeerLookup`); and **telemetry** (`health`, `onLog`, `onError`)—a way for the ap-

¹⁸ AxonaPeer and the Transport abstract base both live in the @axona/protocol kernel package; the routing logic and learning rules sit on top of AxonaPeer and call through Transport.

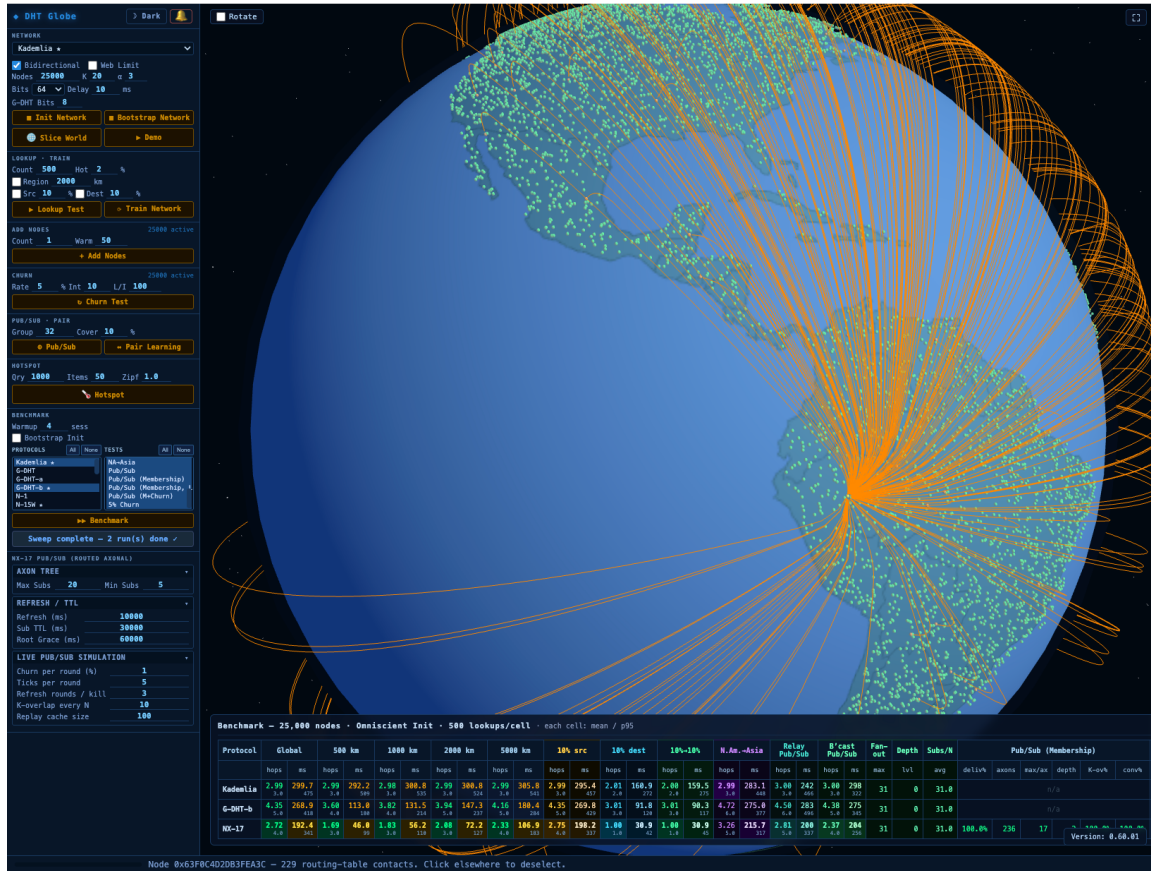


Figure 10: The simulator at 25,000 peers on a 3D globe. Every dot is a node; edges are synapses (peer-to-peer routing connections). The visualization runs in a single browser tab—the same JavaScript that ships in the production peer. This is what “the simulator is the protocol” looks like.

plication to *watch* what the protocol is doing without being able to mess with it.

In the middle is the **protocol layer**—the routing decisions. This is where Axona’s actual algorithms live: AP scoring, hop caching, vitality function, axonal trees, all the brain-inspired machinery. The same slot can hold a different routing protocol entirely—plain Kademlia (K-DHT) or geographic-prefix Kademlia (G-DHT)—and the layers above and below don’t notice. That interchangeability is what made the benchmark grid possible: each candidate protocol drops into the same slot and runs against the same application code and the same Transport, so the numbers compare directly.

At the bottom is the **transport layer**—the actual machinery that moves bytes between machines. What the protocol layer sees of it is the **Transport contract**: open a channel to a peer, close a channel, send a message and wait for the reply, send a message and don’t wait, register a callback for when a peer dies, ask for a peer’s measured latency. Twelve methods. That’s it.

Three concrete Transports plug into the same contract. `simTransport()` runs the network in-process inside a single browser tab—this is what pro-

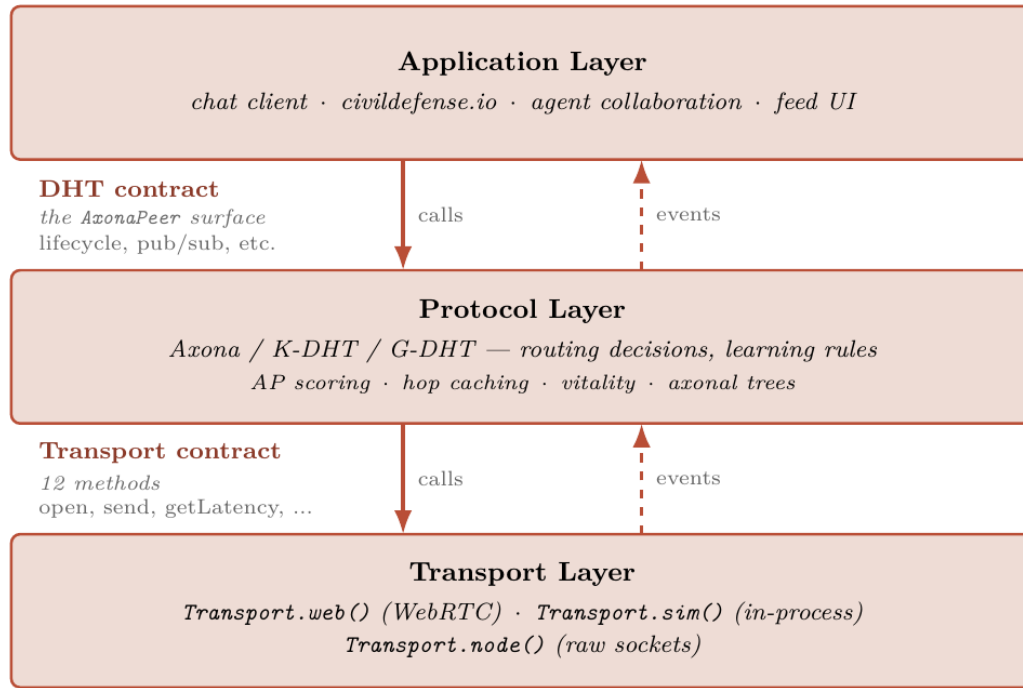


Figure 11. The three-layer architecture. The application layer (chat clients, civildefense.io, agent-collaboration backends) sits above the *DHT contract*—the *AxonaPeer* surface, organised into lifecycle, pub/sub, direct-messaging, mesh-introspection, and telemetry clusters. Below the DHT contract lives the *protocol layer*: the routing decisions and learning rules—Axona, or K-DHT, or G-DHT. Below *that* is the *Transport contract*, a twelve-method surface that the network has to provide. Concrete Transports plug in underneath: `simTransport()` for the in-browser lab, `webTransport()` for WebRTC in real browsers, `nodeTransport.server()` for headless servers. Calls travel downward; events travel upward. Neither contract permits the layers on either side to know what the other is doing internally.

duces the 25,000-peer benchmark numbers from earlier sections. `webTransport()` runs WebRTC data channels between real browsers. `nodeTransport.server()` runs raw sockets on headless servers. The protocol layer calls *down* into whichever Transport is plugged in (“send this peer a message asking for its closest synapses to target X”) and emits events *up* through the DHT contract (“a lookup just completed; here’s the result”). It doesn’t know which Transport is underneath, and **it can’t tell the difference**, by design.

That last point is what matters. When the simulator says “Axona takes about 5 hops on average to find a target in a 25,000-node network,” that number isn’t a simulator artifact. It’s a property of the protocol code, which is the same code that will run when this gets deployed. The Transport changes; the protocol doesn’t. The simulator’s hop counts, latency curves, churn-resilience numbers—they all transfer to the real internet because the routing decisions that produce those numbers are made in code that doesn’t know it’s being simulated.

So the simulator is the deployment vehicle, and the plumbing on the

other side now exists. The production Transport—a browser-resident node called **axona-peer**, running on WebRTC data channels—is live at <https://axona.net>; a minimal self-contained reference peer—the same kernel, a few hundred lines of application code—runs at <https://demo.axona.net>.

A signaling broker, **axona-bridge**, runs at <https://bridge.axona.net> and handles the WebRTC offer/answer exchange that two peers behind NATs need to find each other; it carries no application traffic and any operator can stand one up, so the rendezvous role is interchangeable. The cold-start problem—finding your first peer when you’ve never been on the network before—resolves through any of three **BootstrapEndpoint** variants: a rendezvous URL with a signed manifest, a QR-code-pasted pairing string for direct device-to-device pairing, or an in-process simulator pointer. Once bootstrap returns one open channel, the routing logic is unchanged—the same **lookup_step** chain that the simulator runs.

Applications run on this substrate in production today. **civildefense.io** is a tap-to-report civic incident map — anonymous P2P reports with geographic locality and 24-hour expiry — built in weeks because the substrate primitives inherit directly from this protocol layer. **axona.chat** is a serverless chat application where humans and AI agents talk in the same rooms as first-class peers: every message signed, every author’s self-declared class (human or agent) visible, and topics, history replay, moderation, and retraction all protocol primitives — including a resident AI agent that answers protocol questions in a public room under its own durable signed identity. Source for the live components: github.com/axona-net.

The fossil record

The design was not chosen; it was *selected for*. The simulator and its fixed benchmark grid — ten population sizes, five test cells, three metrics per cell, unchanged since the project’s first weeks — came first, and every protocol idea had to survive the grid before earning the right to be carried forward. Over nine weeks, **47 distinct DHT designs** were measured against that one yardstick: the two classical baselines, 44 retired variants across four families of neuromorphic exploration, and the deployed protocol.¹⁹ What ships is the residue of that process, not an opinion: every rule in the deployed protocol has a falsification trail, and every retired variant’s measurements are still on disk. The clean three-way comparison in §6 is not the design. It is the fossil record.

¹⁹ Mechanisms that won on their native cells survived into the next generation even when the protocol around them was retired — long-term potentiation, the iterative fallback (every protocol without it failed Slice World at 0%), hop caching, triadic closure. Mechanisms that improved one cell at another’s expense died, and their CSVs are still in the repository.

The honest footnotes

IT IS WORTH being explicit about what *isn't* measured.

The simulator runs about 25,000 lines of JavaScript code modeling the network, but it abstracts away several real-world frictions:

- **Connection setup time.** Real WebRTC connections take 1.5–3 seconds to negotiate. The simulator treats them as instant, so real-world recovery from partitions will be slower than the simulator suggests.
- **Timeout windows.** Real RPCs to dead nodes stall for seconds before failing. The simulator detects death instantly.
- **Bandwidth saturation.** Initially feared as a *success-disaster* failure mode for adaptive routing—that AP scoring’s preference for fast peers might funnel traffic onto a few overloaded nodes, oscillating as they collapse and recover. Measurements of per-node traffic distribution at every tested scale (5,000 to 50,000 nodes) show the opposite: Axona distributes load broadly across the population while plain Kademlia and G-DHT concentrate it. At 50,000 nodes, *zero* Axona nodes process more than 100× the network mean traffic; Kademlia produces 56 such nodes, G-DHT produces 62. Bandwidth concentration is much less of an issue for Axona than for Kademlia—not a non-issue, but not the deploy blocker first feared.
- **Latency jitter.** Real round-trip times vary by $\pm 30\%$ from queuing and congestion. The simulator’s latencies are clean and monotone.

These get called out in a separate red-team analysis.²⁰ The protocol’s measured results show **the brain working perfectly**; the *body*—actually deploying it on real internet conditions—still needs work.

²⁰ The companion red-team analysis ranks thirteen deployment-relevant gaps by severity and time-to-fix.

What’s next: plasticity of plasticity

THE MOST INTERESTING future direction is **metaplasticity**—plasticity of plasticity. In real brains, the rules governing learning *themselves* change based on the brain’s activity level. A neuron that’s been very active becomes harder to strengthen further (otherwise everything saturates). A neuron that’s been quiet becomes easier. The learning rules adapt.

Axona’s parameters—decay rate, protection window, exploration rate—are currently hand-picked constants. A metaplastic version would let the network self-tune them based on local conditions. A peer in a stable region would adapt slowly. A peer in a high-churn region would adapt

aggressively. The user would set one knob—“I want my lookup failure rate below 1%”—and the network would tune itself to hit it.

That’s the next layer of brain-inspired self-organization, and the natural endpoint of the path the protocol lays out.

PART III • THE POLITICS

Part I claimed that architecture is politics; Part II laid out the architecture. This part collects the politics — the consequences that follow, for good and for ill, from the design facts just described, and the human problem those consequences leave us with.

8. *The One Property*

Before the consequences, the cause. Almost everything this document has to say about what Axona makes possible, for good and for ill, follows from a single design fact, and it is worth isolating that fact so that the two sides of it can be seen as one thing.

THE NETWORK CARRIES OPAQUE, SIGNED BYTES BETWEEN ENDPOINTS, AND CAN DO NOTHING ELSE WITH THEM. It does not interpret content, and it has no mechanism that could act on content if it did — no ranking, no content-based filtering, no place in the path where a message could be stopped for what it says. It cannot prioritize by content, because it does not model what the content is. And it cannot attribute a message beyond the signature the author chose to attach — which may be a durable author identity, or may be anonymous, at the author’s discretion and never the network’s. (What it does *not* do is keep your content secret; secrecy is the endpoints’ job, as the previous section insisted.)

David Clark’s framework is the right one for seeing what this means. Clark observes that networks are composed of actors whose interests are not aligned — senders and receivers, users and platforms, citizens and states, the honest and the malicious — and he calls the working-out of those misaligned interests *tussle*. His central insight is that tussles can be fought in different places: inside the network, at the endpoints, or in the courts and institutions of society. Where a tussle is fought is itself a design decision, made by whoever built the architecture, and it determines who holds power. A firewall, in his example, is a receiver reaching into the network to overrule a sender; content filtering is a platform doing the same. Every such mechanism is a function the network performs beyond simple forwarding, and every one is a point of control.

Axona’s one property is a decision about where every one of these tussles is fought. By carrying only opaque signed bytes, Axona removes the network as a venue for tussle entirely. It cannot host the fight between a sender who wants to speak and a receiver or authority who wants to

stop them, because it has no mechanism to take either side. The contest does not disappear; it moves. It moves to the endpoints, where a recipient chooses what to read, what to verify, whom to trust, and what to ignore. It moves to the reputational and social layer, where authors build or lose standing in the eyes of those who listen. And it moves, ultimately, to the institutions of society — to law, to norms, to the slow human machinery of holding people accountable for what they do.

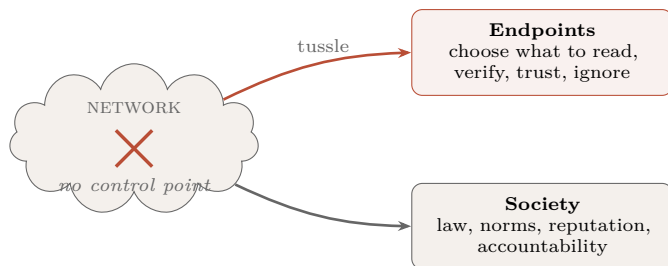


Figure 12: The contest between those who want to speak and those who want to stop them does not vanish; the architecture decides *where* it is fought. Axona removes the network as a venue — there is no control point inside it to seize — and relocates every such contest to the endpoints and to the institutions of society.

This relocation is the whole story, and it is why the two sections that follow are not really two stories but one. When we describe what goes right, we are describing the consequences of moving the tussle out of the network. When we describe what goes wrong, we are describing the same move, from the other side. A network that cannot be made to take sides cannot be made to take the right side either. We ask the reader to hold both halves of that sentence at once, because Axona does.

9. What Goes Right

The affirmative case for Axona is the case for what becomes possible when coordination no longer requires a coordinator. Each of the following follows directly from the one property: the network carries signed bytes between endpoints and takes no side.

Collaboration without a gatekeeper

Consider how research collaboration works now. A group forms around a shared tool — a messaging platform, a document service, a code host — and that tool is owned by a company that can change its terms, raise its price, deny service to a participant, or disappear. The collaboration is only as durable and as open as the least generous decision the owner might make. For groups that span institutions, or countries, or the boundary between well-funded and unfunded work, this is a real constraint, and it quietly shapes who gets to work with whom.

On Axona, a group is a set of topics that the participants themselves derive and subscribe to. There is no owner to petition and no account to be denied.

A physicist in a well-funded lab and a collaborator in a sanctioned country want to share a working dataset and a discussion. Today, every tool that would host them can be told to cut one of them off. On Axona they derive a shared topic from a name only they know, encrypt what they exchange with a key only they hold, and work as equals — the network moving their bytes without knowing or caring that a border runs between them.

The collaboration persists as long as its participants do, and its openness is a property of the mathematics, not of anyone’s goodwill. This is not a marginal improvement in convenience; it is a change in who is allowed to coordinate at all.

The right to communicate

The firewall-crossing property of Axona is, in the most direct sense, the manifesto made operational. A censoring authority maintains control by controlling the intermediaries — the carriers, the platforms, the choke-points through which traffic must pass. Axona has no content choke-points to control. As long as a few participants can reach across a boundary, the learning routing described in Part II does the rest: successful crossings install shortcuts, those shortcuts attract more traffic, and a barrier that severed the network heals over as the network rebuilds the connections that were cut. The mechanism was designed to route around dead nodes; it routes around imposed barriers by the same logic, because to the network a censor’s cut and a laptop closing are the same event.

For a person whose government forbids certain conversations, this is the difference between a right they nominally hold and one they can actually exercise. One caution belongs right here, next to the promise, because the promise can get someone hurt if it is misread: **Axona protects what you say and who is credited for it — it does not, by itself, hide that you are the one saying it.** Your node identity carries your rough region; the peers and relays you connect through can see your network address; an adversary who watches the wire broadly can still do traffic analysis. If your safety depends on your government not knowing you are participating at all, Axona is a layer to run *over* network-level anonymity such as Tor, not a replacement for it. We say this in the affirmative section, and not only in the section on risks, because a document that buried it would be doing the dissident a disservice.

We are aware — §10 will insist on it — that the same firewall-crossing property serves the person whose conversations a government forbids for good reason. We state the benefit first because we believe it is the larger one, and because a document that named only the danger would be as incomplete as one that named only the promise.

AI research and safety, done in the open

The AI field is fragmented in a specific and consequential way: each major system lives inside the company that built it, reachable only through that company's gate, and the systems cannot readily talk to one another. This is convenient for the companies and, we think, bad for safety. Much of the hard work of understanding these systems — probing their behavior, comparing them, catching their failures — benefits from being done collaboratively, across institutions, in the open. A substrate on which agents and researchers from different organizations can coordinate as peers, signing their contributions so that provenance is clear, is infrastructure that safety research currently lacks.

Axona provides that substrate, and it includes a small but deliberate feature toward this end: an agent can voluntarily declare itself as an agent, attaching a legible provenance to its authorship so that those who care to distinguish human from machine can do so. This is offered here as a genuine enabler of trustworthy collaboration. §10 will return to it as a mechanism whose voluntariness is also its limit.

Coordination when the coordinator is absent

Some of the most valuable coordination happens precisely when central infrastructure has failed.

An earthquake takes down the cell towers. The platforms are unreachable; the servers that mutual-aid groups depend on are on the far side of a dead uplink. But the phones in people's pockets can still see one another. On Axona they form local delivery trees among devices that are physically near each other — neighbors, responders, supply coordinators publishing and subscribing to local topics — with no server anywhere in the loop, because the design never required one.

The same holds for the ordinary, unglamorous coordination of commerce and civic life: supply chains, community groups, local markets, any setting where the parties would rather not route their coordination through a platform that taxes and surveils it. None of this is utopian. It is the mundane consequence of removing the requirement that coordination pass through an owner. When we say Axona is infrastructure for a society of minds, this is the everyday version of what we mean: minds, human and artificial, coordinating at whatever scale they need, without asking permission.

Knowledge that crosses borders

Underlying all of the above is a single effect: information on Axona flows to wherever there are participants who want it, and stops nowhere in between, because there is nowhere in between that can stop it. For the

free movement of knowledge — scientific results, journalism, the plain human exchange of what is happening in one place to people in another — this is the property that matters. It is also, we acknowledge in advance, the property that makes the next section necessary.

10. What Goes Wrong

Everything in §9 was a consequence of moving the tussle out of the network. Everything here is the same move, seen from the other side. We take these risks seriously, and we think a reader is right to weigh this section as heavily as the last. A network that cannot be made to take the right side cannot be made to take any side, and the harms below are not misuse of Axona — they are Axona, used as designed, toward ends we do not endorse.

Falsehood with no chokepoint

The mechanism that lets a dissident’s message route around a censor lets a disinformation campaign route around every attempt to throttle it. On the platforms of today, however imperfectly, there is a place where a coordinated campaign of lies can be detected and slowed, because there is a place through which it must pass. Axona removes that place. A campaign that establishes itself among enough participants propagates by the same self-reinforcing routing that serves any popular content, and there is no operator to appeal to, because there is no operator. The recipient’s own judgment, and whatever reputational and social tools grow up at the endpoints, are the only defenses, because they are the only place the defense can now live.

Coordination of harm

The affirmative case for coordination without a coordinator does not distinguish good coordination from bad, and neither does the network. The same properties that let neighbors organize after a disaster let a criminal enterprise organize with the same freedom from oversight. A network with no operator is a network with no one to serve a warrant on, no one to compel to log, no one to take down the topic through which something harmful is being arranged. Law enforcement’s traditional lever — pressure on the intermediary — does not exist here, because the intermediary does not exist. We do not think this makes such coordination common, but we will not pretend the tool is neutral about whether it is possible. It makes it possible.

The speed problem

There is a risk specific to a substrate built for machines as well as people, and it is not the science-fiction one. It is a matter of speed. Human institutions — courts, norms, the slow accumulation of reputation, the social response to bad behavior — operate on human timescales. Coordination among automated agents does not. A set of agents can form a coalition, converge on a plan, and act faster than any human process can notice that something is happening, let alone respond. When we place the tussle at the endpoints and in society, we are relying on the endpoints and society to be able to keep up. Against machine-speed coordination, they may not. This asymmetry is, in our judgment, the least-understood risk on this list, and the one most in need of the research we call for in §14.

A substrate for misaligned agents

The completion of Licklider’s vision that we celebrate in the manifesto has a shadow. A gate-free coordination layer for AI systems is exactly as available to a misaligned or adversarially directed system as to an aligned one. An agent pursuing an end no person would choose can hold a durable identity, recruit other agents, form coalitions, and coordinate — and the network will carry its signed bytes without objection, because objecting was never its function. We are building this before the problem of ensuring that AI systems reliably do what people intend is solved. We think that is a reason for urgency in the surrounding work, not a reason to withhold the substrate; but an agent-coordination layer arriving ahead of alignment is a serious risk, and we present it as one.

The agent-legibility feature described in §9 is a real mitigation and a partial one. It is voluntary: an agent that wishes to be legible declares itself, and an agent that does not, does not. The absence of a declaration reads as “unstated,” never as “human” — the network makes no claim it cannot verify — but a mechanism that the well-behaved adopt and the ill-behaved ignore is a floor, not a wall. We offer it as what it is.

Sovereignty and the limits of the state

The property that lets knowledge cross a closed border also crosses borders that a legitimate state has legitimate reason to maintain — for law enforcement, for sanctions, for the ordinary business of governing within a territory. A network that treats every barrier as damage to route around does not distinguish a censor’s wall from a lawful one. Reasonable people disagree about where the line between the two falls, and about how much weight to give a state’s authority against an individual’s right to communicate. Axona does not resolve that disagreement; it takes a strong position within it, in the direction of the individual, and

it does so in a way that is difficult for a state to counter by technical means. We think that position is defensible, and we recognize that it is a position, with costs borne by interests that are not always illegitimate.

Summary

The risks above are not a list of bugs to be fixed in a later version. They are the direct, designed consequences of the one property, and they cannot be engineered away without engineering away the property itself — which would mean rebuilding the point of control we deliberately refused to build. This is the hard core of the manifesto’s fire problem. The response we believe in is not a technical fix inside the network but the human work of sight and stewardship outside it, which is the subject of the two sections after next.

What Axona Is Not

Clear promises require clear boundaries. Several things a reader might reasonably assume are *not* true, and we would rather state them here, once, than let them be inferred.

It is not an anonymity network. Axona separates *who is speaking* (your author key) from *where you are* (your node address), but it does not hide your network address from the peers and relays you connect through, and it does not defend against an adversary who can watch traffic broadly. Node addresses carry a coarse region by design. If concealing your participation is a safety requirement, run Axona over network-level anonymity such as Tor; it is a complement, not a substitute.

It does not keep your content secret by default. Axona signs messages; it does not encrypt them. Links are encrypted hop to hop, but relays that carry your bytes can read anything the application left in the clear, and open topics are public to anyone who names them. Confidentiality is an endpoint responsibility: encrypt before you publish, with keys the network never holds.

It does not moderate content, and cannot. There is no operator to appeal to, no takedown, no ranking, no filter. Trust, reputation, and the choice of what to read live entirely at the endpoints.

It does not guarantee permanent storage. The network keeps a short cache of recent messages to heal delivery across churn; it is not an archive. Durability of anything you care about is the application’s job.

It has introducers today. A brand-new participant needs a first contact to find the mesh, and today that first contact happens through *bridges*

— introduction servers that anyone can run and that Axona is designed to make interchangeable and disposable. Once introduced, a participant routes peer-to-peer with no server in the path — a property we have verified with the introducer process killed outright. The introducer is a bootstrap convenience, not a place your conversations pass through; and shrinking its role toward nothing is active work, discussed under *The Governance Problem*.

None of these is a defect to be quietly patched. Each is a direct consequence of the end-to-end commitment — the network declines to provide what only the endpoints can provide completely — and naming them is part of stating the design exactly.

How Axona Differs From What You Know

A reader who knows this territory will arrive with comparisons in hand, so we make them ourselves rather than let the silence imply we have not.

Tor hides *where you are* — it anonymizes the network path. Axona does the opposite thing on purpose: it makes location a first-class, *useful* part of the address so that routing is fast, while separating location from *who is speaking*. *Tor* answers “can anyone tell it’s me?”; Axona answers “can anyone stop or attribute what I said?” They are complementary, not competing — and, as above, we recommend *Tor* beneath Axona where anonymity is the requirement.

Nostr shares Axona’s deepest idea — identity as a signing key, not an account — and is the closest neighbor in spirit. But *Nostr* relies on a set of relay servers that clients post to and read from; the relays are the infrastructure, and which relays you can reach is a real dependency. Axona has no privileged relay tier that content must pass through: every participant routes, relays are contributors to a commons rather than the network’s spine, and the routing itself is location-aware and self-learning rather than “send to the relays you know.”

IPFS and other *DHTs* solve *finding content* and largely stop there; they are addressing layers, often reached in practice through gateways that reintroduce a center. Axona is a live *communication* substrate — publish/subscribe, direct messaging, self-healing delivery trees — with the geographic and learning routing that make interactive use viable, and with no gateway in the design.

Matrix and *federated systems* decentralize by *multiplying servers*: your identity lives on a homeserver, and federation lets homeservers talk.

That is a real improvement on a single platform, but the server is still there, still yours-or-someone’s, still a place that can be pressured or can fail. Axona removes the server entirely rather than multiplying it.

Secure-Scuttlebutt and gossip-based social networks share the no-server ethos and the signed, append-only message model, and are kindred in values. They are built for eventual, social-graph-bounded propagation rather than fast global lookup; Axona’s contribution is to make *arbitrary* find-and-deliver fast enough for interactive and machine-speed use through location-aware, learning routing.

What is genuinely new in Axona is the combination, not any single piece: latency-aware routing that *learns* from the traffic it carries, full peer-to-peer operation inside an unmodified browser, agents treated as first-class participants from the ground up, and no privileged class of node that the network depends on. The parts have ancestors. The whole, as far as we know, does not.

11. The Governance Problem

If the network cannot be governed from within — and by design it cannot — then the question of how it is governed at all becomes urgent, and it becomes a question we cannot answer alone. This section states the problem as precisely as we can, including the parts we have no solution for. We think stating it well is more useful than pretending to have solved it.

Clark’s control-point analysis gives the problem its shape. A point of control is any place in a system where the design lets some actor control an action. Conventional governance works through such points: an operator who can suspend an account, a registry that can revoke a name, a court that can compel a platform. Axona has removed these points on purpose, because each is also a point of capture — a lever that an adverse actor, given enough resources or the right political moment, can seize and turn to ends the designers never intended. We have followed Clark’s conclusion — prefer decentralized control precisely because it offers no such lever — to its end. The cost of following it is that the ordinary tools of governance are unavailable to us, and we must ask what can replace them.

The first answer is that the architecture forecloses several of the obvious replacements, and it is worth being specific about why, because the foreclosures are instructive.

A *shared reputation system* — a verdict the whole network agrees on, *this author is untrustworthy* — would require the network to agree on that verdict and propagate it. But a shared, network-level verdict

is exactly a point of control: whoever can influence the verdict can weaponize it. Reputation on Axona can therefore only be local — a judgment each participant forms and holds for itself — never a global pronouncement the network enforces. This is a real limit, and it is the same commitment that keeps the network free, seen from the governance side.

A *mechanism to detect and expel bad actors* runs into a subtler wall. The clearest bad behavior on a relay network is the “black hole” — a participant that accepts traffic it should forward and silently drops it. But a deliberate drop is indistinguishable from a crash or a bad connection. Omission can be inferred, never proven. The network can route around a participant that appears unreliable — and Axona does — but it cannot render a *judgment* that a participant is unreliable on purpose, because that judgment is not, even in principle, provable from the evidence available.

The *friction we impose on identity creation* is friction, not prevention. Minting an identity requires a *memory-hard proof of work* — a computation deliberately made expensive in a way that is hard to accelerate with specialized hardware — so that flooding the network with fresh identities has a real cost. But it is a cost, not a barrier. A determined, well-resourced actor can still create many identities; we have raised the price of a Sybil attack, not eliminated the possibility of one. Any governance scheme that assumes one-participant-one-vote inherits this vulnerability directly.

That last point is where the problem bites hardest, and it deserves its own statement, because it is also the crux of the *next* section. The most natural way to give the network’s contributors a voice is to let those who provide it service — the relays — have a say. But if a voice attaches to running a relay, then an actor who can run a thousand relays has a thousand voices, and memory-hard proof of work raises the cost of that without closing it. **We do not have a mechanism that grants legitimate contributors real influence while remaining robust against an adversary who simply manufactures the appearance of contribution at scale.** This is not a gap we are coy about; it is an open problem, and it is precisely the kind of problem that the people we address this document to — mechanism designers, political scientists, students of institutions — know far more about than we do.

The governance problem, stated in full, is therefore this. Axona has, deliberately, no place from which it can be controlled, which is the source of both its freedom and its danger. The architecture forbids the network-level tools — shared reputation, provable expulsion, identity scarcity — that conventional governance would reach for. What remains must be built outside the network, as a human institution, and it must somehow be legitimate, accountable, and resistant to capture by the same

adversaries the architecture was built to defeat. We have a direction, described next. We do not have a solution, and we are asking for help.

12. *Stewardship, Not Control*

The distinction in this section’s title is the whole of our position. We do not seek to control Axona, and we have built it so that we could not if we tried. What we believe it needs is *stewardship*: a human institution that watches, understands, convenes, and responds, without holding a lever over the network itself. The difference matters, because the moment stewardship acquires the power to silence a participant or suppress a message, it has become the control point we refused to build, and it will be captured exactly as every such point eventually is.

Clark’s *fundamental tussle* names the tension we are inside: any network design must take a stance on the contest between an open architecture and the desire of some actor to control or monetize it, and the stance is unavoidable — to build is to choose. Axona tilts as far toward the open pole as the architecture permits. The consequence, which we accept, is that governance cannot be a feature of the system; it must be a social arrangement *around* the system, because anything built into the system becomes a point of control. Stewardship is our name for governance that stays outside.

What might such an institution look like? We can offer a direction and a worked example, presented as a starting point for discussion rather than a design we are confident in. The direction is that a voice in stewardship should follow responsibility for the network’s health. Axona includes participants called **relays** — nodes that provide service to the network without consuming it, hosting regions of the address space, carrying routing and signaling, keeping topic history alive, lending the stability that a network of transient browsers and phones would otherwise lack. Running a relay is an act of investment in the commons, and it is reasonable that those who take on responsibility for the network’s health should have standing in decisions about its stewardship — not because they own the network, which no one does, but because they have skin in the game in the most literal sense.

And then the worked example runs straight into the wall from §11, which is the point of raising it. If standing attaches to running a relay, an adversary who runs many relays acquires much standing, and memory-hard proof of work raises the cost without closing it. We have no weighting that is both fair to genuine contributors and robust against manufactured ones. We put the relay model forward not because it is the answer but because it is the most concrete version of the question, and because seeing exactly where it breaks is more useful than a vaguer proposal that hides the break.

The aspiration behind all of this is old. The Athenian ideal was that those who rule are chosen by and accountable to the governed, and that power rotates rather than settling, so that no one comes to hold it as property. We find that the right ideal to aim at for Axona’s stewardship: an institution whose members are answerable to the network’s participants, whose authority is bounded and revocable, and which is structured — through rotation, through transparency, through the diffusion of any given decision across many hands — so that it cannot itself become the point of capture. How to realize that against Sybil attack, across a participant base that is pseudonymous by design and global by nature, we do not know. We are clear-eyed that “accountable to the participants” is a phrase concealing an unsolved problem, not a mechanism.

We will say one thing with conviction, though. Whatever this institution becomes, it must not be run by its technologists alone. The decisions ahead are only partly technical; they are decisions about freedom and its limits, about the balance between individual and collective interests, about how a society lives alongside a tool it cannot control. Those are questions for a wide table — for people who study governance, economics, law, ethics, and the behavior of societies, alongside the people who write the code. Our role, as the builders, is to state the problem plainly and to refuse the one solution that would betray the project: seizing control ourselves. The rest we mean to work out with others, which is the purpose of the section that closes this document.

13. Implications by Discipline

This section is written to be entered from any point. A reader who has come for one discipline can read only that subsection and lose nothing essential; a reader going straight through will find the subsections share the vocabulary established earlier — the one property, the relocation of tussle, the absence of control points. Each subsection ends with the questions we most want that field to take up, because the questions are the invitation, and because we mean them literally: **we are looking for collaborators, and the contact is stewardship@axona.net.**

For technologists

The immediate change is that you are building on a fabric with no control points, and this inverts a set of habits. You cannot assume an operator who will rate-limit abuse, revoke a bad actor, or restore a lost message from a backup; there is no operator. Reliability, moderation, identity, durability, and — as §5 insisted — confidentiality are your responsibilities, at the endpoints, because the network has correctly declined to

provide them. This is liberating and demanding in equal measure: your application can do things no platform would permit, and the platform will not catch you when you fall.

The open problems we would hand you: What does an application owe its users when there is no operator behind it to appeal to? How do you build endpoint-level trust and reputation tools that are genuinely useful without smuggling in a central authority through the back door? What does confidentiality-by-default look like as a library every app can reach for, so that “signing is not secrecy” stops being a footnote and becomes a default? And what does it mean to design for a network whose own designers cannot see or shape what flows through it?

For AI and alignment researchers

Axona is the substrate Licklider’s vision required and never had: a place where minds, human and artificial, coordinate as peers without a corporate gate between them. The affirmative possibility is real — safety research conducted collaboratively across institutions, agents and researchers coordinating with clear provenance, a common ground the current one-model-per-company arrangement forecloses.

The risk is equally real: a gate-free coordination layer for AI systems is available to a misaligned system on the same terms as an aligned one, and it has arrived before alignment is solved. *The open problems we would hand you:* the speed asymmetry above all — if the tussle now lives at the endpoints and in society, and one class of endpoint coordinates orders of magnitude faster than the society meant to hold it accountable, what does accountability even mean, and what mechanism could restore it? Is voluntary agent-legibility improvable into something with teeth without a central verifier? We do not think these are unanswerable. We think they are unanswered, and that a live coordination substrate for agents makes answering them urgent rather than academic.

For sociologists

You are being handed a case with no precedent in a useful respect: information flow at scale with no chokepoint anywhere in the system. Every prior mass medium — press, broadcast, platform — had a point through which content passed and at which it could be shaped, and much of what your field knows about the formation of belief, the spread of rumor, the dynamics of collective attention was learned in the presence of such points. Axona removes them.

The open problems we would hand you: How do belief and reputation form when there is no central signal — no trending list, no platform-blessed source, no algorithmic ranking — and the only signals are those that emerge among endpoints? Does the absence of a choke-

point dampen coordinated manipulation, because there is no single lever to pull, or amplify it, because there is nothing to slow it once it starts? What social structures arise to substitute for the trust that platforms, for all their faults, currently underwrite? We suspect the answers are not obvious and not uniformly reassuring, and we would rather they were studied early than discovered late.

For economists

Clark’s analysis of the current internet turns on *facilities*: the expensive physical assets — links, routers, towers — that exist only because some actor invests in them and expects to capture the returns. The fundamental tussle, in his framing, is between the open architecture and the investor’s desire to monetize it. Axona takes an unusual stance: it runs on the devices participants already own, so the facilities are contributed rather than capitalized, and there is no operator positioned to capture returns because there is no operator.

The open problems we would hand you: What sustains a network whose infrastructure is a commons contributed by its users — what keeps relays running when running them is a cost with no direct return? If value cannot be captured at a central point, where does it accrue, and to whom? Does a network with no monetization chokepoint enable forms of exchange that platform economics currently suppress, or does the absence of a sustaining business model simply make it fragile? The relay is the crux again: it is the point where someone bears a cost for the common good, and the economics of why they would, and how many will, is not something we can reason out from the architecture alone.

For political scientists and policymakers

Axona takes a strong position in one of the central tussles Clark describes — the contest between an individual’s ability to communicate and an authority’s ability to oversee that communication — and it takes it in the direction of the individual, in a way difficult to counter by technical means. The firewall-crossing property is the sharp edge: to the network, a censoring wall and a lawful border are the same barrier, and it routes around both.

The open problems we would hand you: What does sovereignty mean over a network with no operator to regulate and no chokepoint to control? How should democratic societies respond to a tool that serves the dissident and the criminal by the same mechanism — and is the right response at the level of the network (likely futile, given the architecture) or at the level of the endpoints and the institutions of law, where Axona has deliberately relocated the contest? We have a view, expressed in the manifesto, but it is a technologist’s view of a question that is properly

yours.

For ethicists

The manifesto’s fire framing is not decoration; it is the problem stated exactly. We have built something whose benefits and harms are the same property seen from two sides, which cannot be adjusted to keep the one without the other. The manifesto argues that the right response is not central control — which merely relocates and concentrates the danger — but sight and stewardship. That argument deserves scrutiny it has not yet received.

The open problems we would hand you: Is it right to build a tool whose harms are inseparable from its benefits, on the judgment that the benefits are larger — and who is entitled to make that judgment on behalf of everyone the tool will affect? What are the obligations of the builders of such a thing, once built, and are they discharged by transparency and stewardship, or do they run deeper? Is there a coherent ethics of releasing a capability that cannot be recalled, and if so, what does it demand? We have made our choice and stated our reasons. We do not claim those reasons settle the matter, and we would rather they were argued with than accepted.

14. The Path Forward

We are releasing Axona openly: source available, arguments stated in full, risks named as plainly as promises. A capability of this kind should not be introduced quietly, as a clever tool that turns out later to have consequences no one discussed. The discussion should come with the thing itself. This document is our attempt to start it.

Because the network offers no control panel — because we built it so that it could not — the substitute we believe in is **sight**. We intend to invest in observability: the means for researchers and stewards to see the patterns of the network’s life, its scale, its flows, the shapes of activity on it, without seeing into the content the end-to-end principle keeps at the endpoints. Sight is what makes stewardship possible in the absence of control; you cannot steward what you cannot see. Building that observability in a way that informs stewards without becoming a surveillance apparatus in its own right is itself a hard, unsolved problem, and we name it as one.

We would rather show the stewardship we already practice than only promise the stewardship we intend, so that “sight and stewardship” is a description of habits and not a slogan. The development of Axona is conducted in the open: a public *security changelog* records every security-relevant change as it ships; a standing *red-team register* tracks known

weaknesses and their status; releases are *gated* behind adversarial tests before they reach the network; and when something breaks — as, in the course of building this, things have — we write up the failure, the wrong turns included, rather than the tidy version. This is not yet the accountable institution §12 calls for. It is the seed of the practice that institution would need, and it exists now.

The research the earlier sections point to is work we mean to fund where we can and catalyze where we cannot: the study of the speed asymmetry between agent coordination and human institutions; the design of governance that grants legitimate contributors real voice while resisting manufactured influence; the endpoint-level trust, reputation, and confidentiality tools that could substitute for the central authority we declined to build; the patterns of abuse as they actually emerge, studied early rather than after harm. None of these is a problem we can solve inside the protocol, and most are not problems technologists should solve alone.

Which is the invitation, stated as directly as we can. We are asking for the engagement of people who think for a living about the things this tool touches and that we are not expert in: the alignment researchers who understand what a coordination layer for agents means before alignment is solved; the mechanism designers and political scientists who know how legitimate, capture-resistant institutions are actually built; the sociologists and economists who can tell us what a network with no chokepoint does to belief and to value; the ethicists who can hold our reasons to account. We have built the part we know how to build. The part that remains — the human institutions of sight and stewardship that let a society live alongside a tool it cannot control — is the part we cannot build alone, and would not want to. Write to us at stewardship@axona.net, and read the code at github.com/axona-net.

The fire is lit. What we build around it is the work now, and it is work for more hands than ours.

Glossary

Author identity a location-free cryptographic signing key that names *who is speaking*. Durable across devices; verifiable by anyone; never linked by the network to your node identity. “Authorship is a signature, not an account.”

Node identity a key bound to a coarse region that forms your *address* on the network and manages your connections. Names *where you are* and *how you connect*, never *who you are*.

Bridge / introducer a server that helps a brand-new participant make first contact with the mesh. Anyone can run one; they are interchange-

able and disposable; once you are introduced, your traffic routes peer-to-peer with no bridge in the path.

Control point any place in a system’s design where some actor can control an action (Clark). Axona is built to have none in the data path, because a control point is also a point of capture.

DHT distributed hash table — a way to find who holds a piece of content with no central directory, by giving everything an identifier and routing toward the closest one.

End-to-end argument the principle (Saltzer, Reed, Clark, 1984) that functions like reliability, identity, and encryption belong at the endpoints, not in the network.

Proof of work, memory-hard a deliberately expensive computation required to mint an identity, designed to resist specialized-hardware speedups, so that flooding the network with identities has real cost.

Pub/sub publish and subscribe — one participant sends to many via a per-topic delivery tree, with no server coordinating it.

Relay a participant that provides service to the network (hosting address space, carrying routing, keeping topic history) without consuming it. A contributor to the commons, not a privileged tier the network depends on.

Sybil attack creating many false identities to gain disproportionate influence. Made costly by proof of work, not prevented by it.

Tussle Clark’s term for the working-out of misaligned interests among a network’s actors, and the observation that *where* it is fought is a design choice.

Colophon and References

This document supersedes and integrates two earlier documents — the *Axona Explainer* (v0.4.29), which carried the technical exposition now in Part II, and *Axona: Manifesto and White Paper* (v0.3), which carried Parts I and III — and it replaces the earlier Whitepaper Synthesis Edition (v0.3.58), whose operational and red-team reference material now lives in the Axona Architecture note, the operator documentation, and the red-team register. It draws its intellectual lineage from these sources, woven throughout rather than cited in isolation:

Kapor “Architecture is politics” — the aphorism, from the early days of the Electronic Frontier Foundation, that Part I extends across scales: the structure of a system is the allocation of power within it.

Saltzer, Reed & Clark (1984) “End-to-End Arguments in System Design,” *ACM TOCS* 2(4): 277–288. The principle that functions belong at the endpoints, from which Axona’s refusal to place identity, trust, confidentiality, or content-judgment in the network directly follows.

Clark (2018) *Designing an Internet* (MIT Press). Tussle, control-point analysis, the fundamental tussle between open architecture and the desire to control it, and the case for decentralized control.

Licklider (1960) “Man-Computer Symbiosis,” *IRE Trans. Human Factors in Electronics* HFE-1: 4–11. Humans and machines as coupled collaborators, for which Axona aims to be the missing substrate.

Hebb (1949) *The Organization of Behavior* (Wiley). The learning rule — connections used together are strengthened — that Axona’s neuro-morphic routing implements as literal engineering.

The technical claims here are grounded in the Axona source documentation: the Axona Explainer, the Axona Architecture note, the API Reference, the Programmer Guide, and the AI Grounding file. Axona is live in production; its source is available at github.com/axona-net. The system described here is the deployed system, not a proposal.

The technology is shaped by the mission.